Same Formula, Different Signal:
Funding-Rate Censoring and the Premium Gap between Hyperliquid and Centralised Perpetuals

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Abstract

Hyperliquid, Binance and Bybit set perpetual funding by one rule: while the averaged premium lies in a 10 bp band, funding equals an interest rate of 0.01% per eight hours, and a settlement reveals the premium only as that band. From 2 October 2025 to 28 April 2026, 46.0% of Hyperliquid's hourly BTC settlements sit at that rate, against 5.9% on Binance and 7.7% on Bybit. From 11 December 2023 to 30 September 2026 the shares are 55.2%, 23.8% and 27.3% for BTC and 52.0%, 23.1% and 28.3% for ETH; taken together, the two quarters in which the exchanges' premiums were positive reverse the ordering. Matched hour by hour with the exchanges' own premium indices, Hyperliquid's BTC premium lies a median 2.2 bps above both (ETH 1.7), in four hours out of five and in every quarter. For BTC the gap persists but varies: 3.8 bps in 2024, 3.0 to 3.1 in 2025, 0.3 to 0.5 in 2026Q1 and 2026Q2 and 1.3 to 1.5 in 2026Q3. By a Shapley split it accounts for about all of the difference in shares from 2024Q3 to 2025Q4, but for less than half in 2026Q1 and 2026Q2 (ETH: 55% to 70%), when all three median premiums lay below the band and dispersion mattered more. Cross-venue comparisons of positioning should therefore use premiums, with their constructions in view, not headline rates. A two-state Markov-switching AR(1) on the premium reverts faster under stress only around October 2025: allowing for switching, the half-life ratio is 1.5, not significantly different from one under a misspecification-robust covariance that one hour of the 10 October crash dominates, and later data reverse it. A three-state Gaussian hidden Markov model separates hourly volatility into 14.6%, 33.3% and 88.4% annualised states and estimates the direct move from the high to the low state at zero to three decimals, as do its fits to simulated single-regime Student-t GARCH paths. All results are descriptive.

Keywords: funding rates; censoring; perpetual futures; premium index; Hyperliquid; Binance; Bybit; regime switching; hidden Markov models.
JEL: C22, C58, G12, G13.

1 Introduction

Perpetual futures are the most traded cryptocurrency derivatives [27], and decentralised venues, Hyperliquid foremost [38], now trade them at scale. Having no expiry, a perpetual is tied to its index by a recurring funding payment between longs and shorts, set from the contract's averaged premium over its index [1, 31]. Hyperliquid, Binance and Bybit compute that payment with one rule, which pays exactly an interest rate whenever the averaged premium lies inside a band 10 bps wide. Inside the band the published rate stops moving with the market. How often a venue's rate sits there decides whether its headline funding can be read as a signal of positioning, and whether funding compared across venues compares like with like.

Funding has a literature on carry, basis and, increasingly, the rule itself. Schmeling et al. [48] trace the large and volatile basis of crypto futures to trend-chasing demand for leverage meeting limited arbitrage capital, Makarov and Schoar [45] document the frictions that segment crypto markets, and Alexander et al. [4] study price discovery on the first large perpetual venue. Werapun et al. [51], Chen et al. [12] and Erez and Smirnov [19] compare funding, its mechanisms and its arbitrage across centralised and decentralised venues, Erez and Smirnov [18] show how the timing of funding shapes no-arbitrage bounds on two decentralised venues, and Lau [38] finds Hyperliquid's funding about 7% a year above Binance's

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and Bybit's over three years. Zhivkov [53], Zhivkov et al. [54] correlate and Granger-test the funding rates of 26 centralised and decentralised venues, Hyperliquid, Binance and Bybit among them, between November 2025 and January 2026, and find funding less correlated between decentralised venues and the exchanges than among the exchanges, a two-tier market in their reading. Pricing models treat the clamp explicitly [27, 30], Kim and Park [36] derive funding rates that keep a perpetual's price anchored to its target, also when the rate is averaged over the past eight hours, and He et al. [28] argue that averaging and the rule's flat and capped regions separate the nominal funding template from the funding delivered at settlement. Counting settlements at the rule's flat value has precedents: He et al. [27] find 41.7% of Binance's BTC settlements at the interest rate over 2020–2024, Ahmed and Bhuyan [3] report a mass at the formula point on centralised exchanges but not on dYdX, whose funding carries no interest component, and in a companion study over February 2025 to February 2026 find Hyperliquid's funding distribution clustering with the exchanges' rather than with dYdX's [2]; Lim [42, 43] counts 73% of Kalshi's settlements exactly at zero, with Bybit's BTC premium, at a mean of about -4.7 bps over five weeks, as the offshore comparison. On the regime side, hidden Markov and Markov-switching models of Bitcoin returns [7, 24, 37, 44] build on Hamilton [26] and Rabiner [46].

We make three contributions, all descriptive. First, we measure the censoring exactly, from the venues' decimal settlements, for a decentralised venue and two exchanges over common windows: the 2 October 2025 to 28 April 2026 window of our regime models, and the whole period since Hyperliquid adopted the exchanges' clamp, 11 December 2023 to 30 September 2026, for BTC and ETH (Section 4). Hyperliquid is censored far more often than either exchange in most months, but not in all: its share fell below both exchanges' in four months for BTC and three for ETH, mostly in 2024, in each of which 41% to 75% of its hours lay above the band; taken together, the two quarters in which the exchanges' premiums were positive reverse the ordering. Second, we measure the premium gap behind that ordering, hour by hour and with each venue's own premium construction, and split the difference in censored shares into the gap's location, the averaging window and the premiums' dispersion (Section 5). Third, we describe the regime dynamics of Hyperliquid's premium, with a misspecification-robust (sandwich) covariance beside the Hessian and switching-adjusted half-lives, and of its returns, and report two cautions that travel: an asymmetry in mean reversion that belongs to one episode, estimated on a likelihood whose most frequent optimum is not its maximum, and a boundary zero in a hidden Markov kernel that a single-regime GARCH process produces as well (Section 6).

Section 2 sets out the rule on the three venues and Section 3 the data; Section 7 draws the implications and states the limitations, and Section 8 concludes.

2 The funding rule on three venues

Hyperliquid's documentation gives the eight-hour funding rate as

$$F_t \;=\; \bar{P}_t + \mathrm{clamp}\big(r - \bar{P}_t,\, -c,\, +c\big), \tag{1}$$

with $\bar{P}_t$ the premium index averaged over the funding interval, $r = 0.01\%$ per eight hours the interest rate, $c = 0.05\%$ the clamp and $\mathrm{clamp}(x, a, b) = \max(a, \min(b, x))$; Hyperliquid pays $F_t/8$ every hour [31]. Binance and Bybit apply the same r and c and settle every eight hours [8, 9]. The fixed 0.01% is an industry convention [5], and Hyperliquid states that it adopts it for consistency with the exchanges.

The clamp is the identity when $|r - \bar{P}_t| \le c$, that is, when $\bar{P}_t$ lies in the band $[r - c,\, r + c] = [-4, +6]$ bps, centred on +1 bp. There $F_t = r$ exactly: the rule pins the rate at the interest rate. Below the band the rate is $\bar{P}_t + c$, below r; above it, $\bar{P}_t - c$, above r. A settlement outside the band therefore identifies the averaged premium exactly, while one at r reveals only that the premium lay in the band. As a measurement of the premium the rate is interval-censored, a case of middle censoring [34] or coarse data [29]. We call such settlements censored and, as a label only, call r the floor: the rate falls below it whenever the premium lies below the band, as it does in most exchange settlements (Section 4).

Hyperliquid has not always used this rule. Its published rates, classified record by record, settle every eight hours until 8 June 2023 and hourly since. Until 11 December 2023 they fit a $\pm 0.03\%$ clamp (a band of -2 to $+4$ bps), with two exceptions: from 16 June to 15 July 2023 they equal the premium alone, and a few records on 16 July 2023 fit a $\pm 0.02\%$ clamp. From the record stamped 11 December 2023 23:00 UTC they fit the exchanges' $\pm 0.05\%$ (Table 1, note a). These rules and dates are inferred from the data; we found no announcement. Under the common rule a Hyperliquid hour is at the floor exactly when its premium lies in the band, with no exception in 24,577 hours for BTC or ETH. Every long-run comparison below starts at the record of 11 December 2023 23:00 UTC.

The rule is common; the premium that enters it is not (Table 1), a point Erez and Smirnov [19] make for perpetual venues in general. Hyperliquid averages premiums sampled every five seconds over one

Table 1. How the three venues build the premium that enters the common rule (1), BTC and ETH perpetuals, from their documentation [8–10, 31–33] and public APIs as retrieved on 8 October 2026; notes a and c record what the published settlements add. IMR and MMR: initial and maintenance margin rates.

	Hyperliquid	Binance (USDT-M)	Bybit (linear)
Premium	$[\max(b-O,0)-\max(O-a,0)]/O$; b, a: average fill prices for the impact notional on the bid and ask; O: oracle	same form against the price index	same form against the index price
Sampling	every 5 s	every 5 s	every minute
Averaging	over the hour (equal weights implied)	8 h, weights $1,\dots,n$ rising to settlement, $n = 5{,}760$; equal weights when settling hourly	8 h, weights $1,\dots,480$ rising to settlement; equal weights until 7 Mar 2024[c]
Reference price	oracle: stake-weighted median of the validators' weighted medians of spot mid prices on Binance, OKX, Bybit, Kraken, KuCoin, Gate, MEXC (weights 3, 2, 2, 1, 1, 1, 1), every 3 s[b]	price index: weighted average of spot prices; BTCUSDT[d]: Binance 0.435, OKX 0.130, Coinbase 0.130, Bitget, Bybit, KuCoin, MEXC 0.065 each, Gate 0.043	index price: weighted average of the six largest spot pairs by 24-h volume; BTCUSDT[d]: Binance 0.339–0.348, Bybit 0.220–0.234, Gate 0.215–0.219, OKX 0.138–0.147, KuCoin 0.070–0.071, Coinbase 0.001
Impact notional	20,000 USDC (BTC, ETH)	200 USDT / initial margin rate at maximum leverage	set per contract in USDT (value not in the pages used)
Interest r	0.01% per 8 h (0.00125% per hour)	0.01% per 8 h	0.01% per 8 h
Clamp c	±0.05% from the record of 11 Dec 2023 23:00 UTC[a]	±0.05%	±0.05%
Cap	4% per hour	±0.75× maintenance margin rate; API: ±0.300% (BTC, ETH)	$\pm 0.75 \times (\text{IMR} - \text{MMR})$, at most ±MMR; API: ±0.333% (BTC, ETH)
Settlement	every hour, $F/8$	every 8 h (00, 08, 16 UTC); hourly while the rate is capped	every 8 h; hourly while the rate is at its limit
Margin, quote	USDC margin; contract and oracle in USDT, whose profit and loss is paid in USDC without conversion (quanto)	USDT margin and settlement; contract in USDT	USDT margin and settlement; contract in USDT
Changes since 2023	8-hourly until 8 Jun 2023; premium only, 16 Jun–15 Jul 2023; ±0.03% until 11 Dec 2023[a]; oracle wording, late 2025[b]	none seen in the settlements from Dec 2023 on	from equal to rising weights, 7 Mar 2024[c]

[a] Inferred from Hyperliquid's published hourly rates; no announcement found. Settlement was every 8 h until 8 June 2023 and hourly since. From 16 June 21:00 to 15 July 02:00 UTC 2023 every rate equals the premium alone ($F = \bar{P}$, paid as $\bar{P}/8$ each hour: no interest, no clamp), and on 16 July a few records fit a ±0.02% clamp instead; otherwise the rates fit ±0.03% (band −2 to +4 bps) up to the record stamped 11 December 2023 22:00 UTC and ±0.05% from the one stamped 23:00.
[b] The oracle page read "spot prices" in an Internet Archive snapshot of 6 October 2025 and "spot mid prices" from one of 10 December 2025 on; whether the method changed is not documented.
[c] Found in the data: Bybit's settlements are reproduced by equal weights through the one of 7 March 2024 08:00 UTC and by rising weights from 16:00; not in the documentation retrieved.
[d] Weights from the public APIs on 8 October 2026: Binance's at 16:29 and 16:52 UTC (identical), Bybit's from three snapshots between 16:29 and 18:29 UTC, printed as the range over the three. Bybit's weights vary over time: they follow 24-hour volumes ("real-time weight", updated hourly according to its help page) and moved between the snapshots; no snapshot need describe the sample period.
Sources, retrieved 8 October 2026: Hyperliquid documentation (Funding, Oracle, Contract specifications); Binance FAQ "Introduction to Binance Futures Funding Rates" (updated 6 March 2026); Bybit Help Center "Introduction to Funding Rate" (22 May 2026) and "Index Price Calculation" (19 June 2026); Binance and Bybit public APIs.

hour; Binance and Bybit average over eight hours with weights rising linearly towards settlement, Bybit with equal weights until 7 March 2024, a switch we find in its settlements. Hyperliquid's reference price is a stake-weighted median of its validators' weighted medians of spot mid prices on seven exchanges (Table 1, note b); the exchanges use weighted-average indices, in which Binance's spot book has the largest BTCUSDT weight on both exchanges (note d). Impact notionals differ, and Hyperliquid margins its USDT-denominated contracts in USDC without conversion. Each of these could shift the averaged premium; we do not identify which does.

Table 2. Summary statistics, Hyperliquid BTC perpetual. Premium: the averaged premium of each hourly funding record, in eight-hour-equivalent bps; funding: the hourly rate. Estimation window: 2 October 2025 to 28 April 2026, $T = 5{,}001$; sub-period: 3 December 2025 to 28 April 2026, $T = 3{,}507$, the estimation-window hours whose hourly prices survive. Skewness without bias correction.

Series	Window	Mean	Std. dev.	Min	Median	Max	Skew
Premium (bps)	estimation	−3.15	2.87	−10.55	−4.11	23.61	1.71
Funding rate (bps/h)	estimation	0.059	0.103	−0.694	0.111	2.327	0.94
Log-return (bps/h)	sub-period	−0.55	50.60	−367.5	−0.14	336.7	−0.26

3 Data

Hyperliquid's two public endpoints are `fundingHistory`, which returns each settlement (eight-hourly until 8 June 2023, hourly since) with the averaged premium it was computed from, for BTC and ETH from 12 May 2023 to the settlement stamped 1 October 2026 00:00 UTC, which pays for the last hour of 30 September; and `candleSnapshot`, which returns candles of traded prices. For Binance and Bybit we take the BTCUSDT and ETHUSDT perpetuals' funding settlements and their one-minute premium-index klines, which we aggregate to the hours Hyperliquid's records cover (Appendix B). Rates are compared as decimals parsed from the venues' strings, so 'at the floor' is an exact comparison. Figure 1 maps the windows and the series.

The regime models use the estimation window, 2 October 2025 00:00 to 28 April 2026 08:00 UTC: 5,001 hourly Hyperliquid settlements and 626 settlements per exchange. The later window runs to 30 September 2026 (3,736 hours, 467 settlements) and the long history from 11 December 2023 23:00 (24,577 hours, 3,073 settlements per exchange). Hyperliquid publishes no record stamped 15 August 2024 13:00 UTC for either asset, so the long history holds one hour fewer than it spans, and the block comparison of Section 4.2 uses the 3,071 settlement blocks with all eight hourly records under the common rule: the first settlement, 12 December 2023 00:00, averages six hours from before the switch, and the block ending 15 August 2024 16:00 lacks that hour (the Shapley split averages Hyperliquid's hours in all 3,073 blocks). Funding is otherwise fully replicable, because `fundingHistory` keeps a contract's whole history; the premium series re-retrieved in August and October 2026 equals the archived panel digit for digit on all 3,507 overlapping hours. Hourly prices are not: `candleSnapshot` serves only the latest 5,000 or so candles at each interval, so the hours before 3 December 2025 had left its hourly window before the panel was archived. The return fit on the whole estimation window (Section 6.2) rests on the original extract, retrieved on 28 April 2026; every other result is rebuilt from shipped data, the return-side ones on the hourly sub-period, 3 December 2025 to 28 April 2026 (3,507 hours). Two-hourly candles cover the whole estimation window and agree with the surviving hourly panel aggregated to two hours on all 1,753 overlapping bars.

Table 2 summarises the BTC series. Over the estimation window Hyperliquid's BTC perpetual traded between hourly closes of \$126,101 (6 October 2025) and \$62,883 (24 February 2026), as the original extract records them. The premium averaged −3.15 bps, the contract trading below its index for most of the window, while funding averaged +0.059 bps per hour, about 5.2% a year, longs paying shorts: in the 46.0% of hours with the premium in the band the contract paid the floor whatever its positioning.

4 Censoring across venues

4.1 The estimation window

Figure 2(a) draws the rule: funding is flat at the floor across the band and moves one for one with the premium outside it. Over the estimation window Binance prints the floor in 37 of 626 settlements (5.9%) and Bybit in 48 (7.7%), against 2,300 of 5,001 hours (46.0%) on Hyperliquid (Table 3). No exchange settlement lies above the floor, and 94.1% of Binance's and 92.3% of Bybit's lie below it. The five months after the window repeat the pattern: 51.7% on Hyperliquid against 10.9% and 7.9%.

The premiums explain the difference (Figure 2(b)). Recovered exactly from the settlements outside the band, the exchanges' premiums are packed just below the band's lower edge, with medians of −4.75 and −4.82 bps and interquartile ranges of 0.66 and 0.60 bps. Hyperliquid's straddles the edge: its median is −4.11 bps, 52.5% of its hours lie below the band, and a long tail into the band lifts its mean to −3.15.

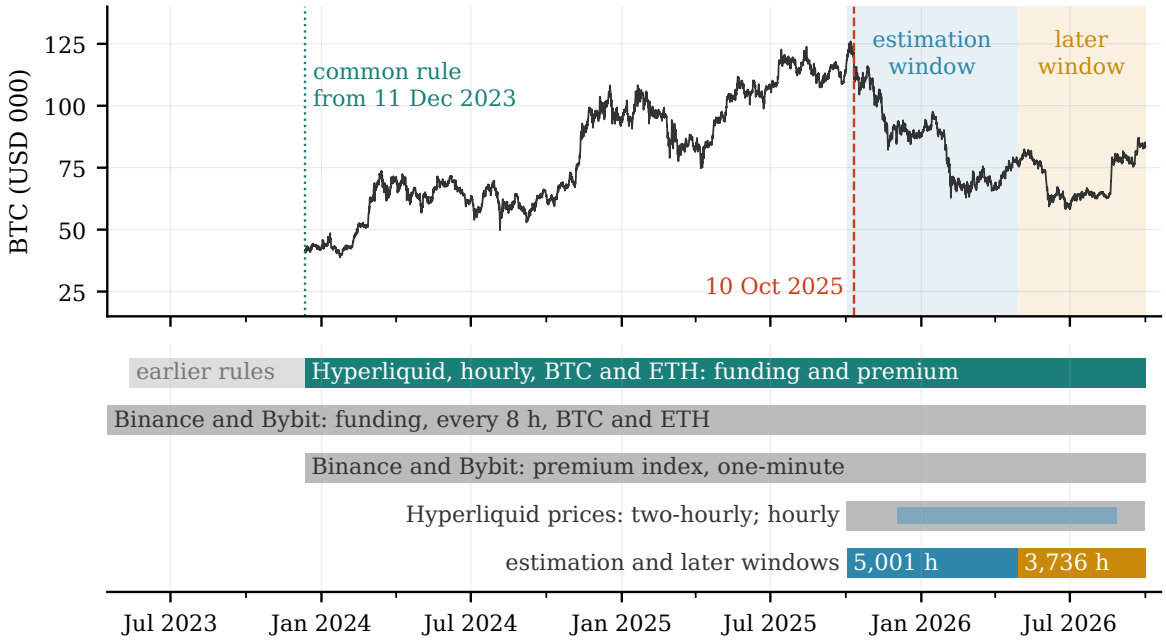


Figure 1. Windows and data. **Top**: the BTCUSDT perpetual on Binance, hourly close, for orientation, with the estimation window (blue) and the later window (ochre) shaded, the record from which Hyperliquid applies the exchanges' clamp (dotted) and the hour of the premium's maximum, 10 October 2025 (dashed). **Bottom**: the series the replication package ships, from first to last stamp: Hyperliquid's funding and premium (grey under its earlier rules), the exchanges' settlements (shipped from 2019 and 2020) and one-minute premium index, and Hyperliquid's two-hourly candles (grey) with the surviving hourly ones (blue) inside them.

Table 3. Funding settlements exactly at the floor, by venue and window, BTC perpetuals, compared as decimals. *Below*: settlements under the floor, the premium below the band. *Median* and *Quartiles*: the premium in bps, Hyperliquid's hourly average and the exchanges' recovered from their settlements; settlements at the floor rank between those below and above the band, so the quantiles are exact. The second row reads Hyperliquid's hourly series at the exchanges' 00, 08 and 16 UTC stamps.

		2 Oct 2025 – 28 Apr 2026					28 Apr – 30 Sep 2026	
Venue	Settles	n	At floor	Below	Median	Quartiles	n	At floor
Hyperliquid	hourly	5,001	46.0%	52.5%	−4.11	[−4.93, −2.26]	3,736	51.7%
Hyperliquid	at 00/08/16 UTC	626	48.4%	50.0%	–	–	467	52.7%
Binance	every 8 h	626	5.9%	94.1%	−4.75	[−5.10, −4.44]	467	10.9%
Bybit	every 8 h	626	7.7%	92.3%	−4.82	[−5.09, −4.49]	467	7.9%

The hourly schedule does not cause the difference. Read every eighth hour, Hyperliquid's series is at the floor in 44.2% to 48.4% of hours depending on the phase, and in 48.4% at the exchanges' 00, 08 and 16 UTC stamps; averaged over the eight hours each exchange settlement pays for, with the exchanges' weights rising towards settlement, its premium lies in the band 46.5% of the time (48.6% with equal weights), against 46.0% of hours. An hourly schedule multiplies censored payments, not the censored share.

4.2 Since December 2023

Figure 3(a,b) follows the monthly shares from January 2024, and Table 4 gives them by year and half-year. Over the whole common-rule period Hyperliquid's BTC funding is at the floor in 55.2% of hours, against 23.8% of Binance's and 27.3% of Bybit's settlements; ETH gives 52.0%, 23.1% and 28.3%. The ordering is not fixed. Hyperliquid's share exceeds both exchanges' in 26 of the 33 full months for either asset, including every month since August 2025, but lies below both in four months for BTC (January, February and November 2024 and July 2025) and three for ETH. In those months Hyperliquid's median premium was positive, +5.0 to +8.4 bps, with 41% to 75% of its hours above the band, in each month a larger share than of either exchange's settlements. The exchanges' median premiums lay in the band, except in November 2024, when three of the four lay above it as well, though below Hyperliquid's.

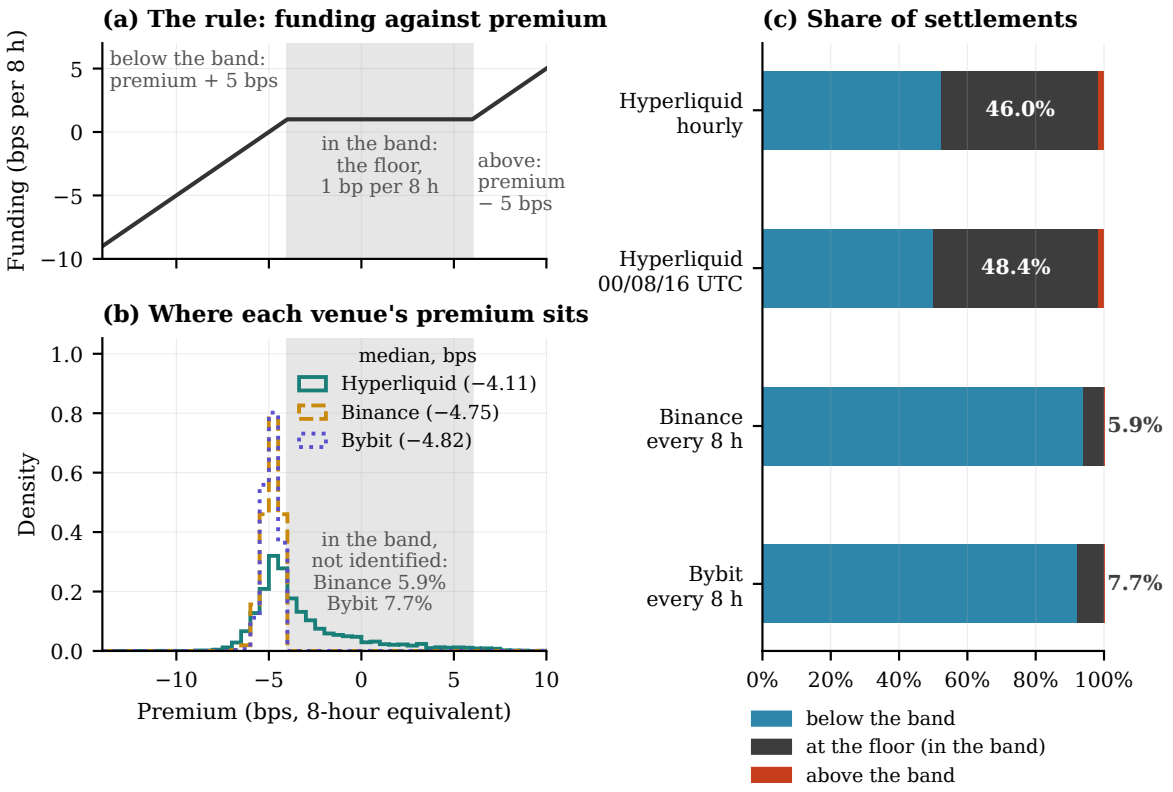


Figure 2. Funding-rate censoring across venues, BTC perpetuals, 2 October 2025 to 28 April 2026. **(a)** The rule (1): funding per eight hours against the averaged premium; inside the band $[-4, +6]$ bps it pays exactly the floor of 1 bp. **(b)** Where each venue's premium sits: Hyperliquid's hourly averaged premium, and the Binance and Bybit premiums recovered exactly from their settlements; settlements at the floor say only that the premium lay in the band and are not drawn, so each exchange's curve integrates to its identified share. **(c)** Shares of settlements below the band, at the floor and above it: Hyperliquid hourly and read at 00, 08 and 16 UTC, and the eight-hourly settlements of Binance and Bybit.

What holds in every quarter, though not in every month, is the order of the premiums. Comparing Hyperliquid's mean premium over each eight-hour settlement block with the premium the exchange's settlement identifies, or with the band when the settlement is censored, Hyperliquid's premium is higher in at least 71.0% and at most 85.6% of the 3,071 blocks against Binance, and in 69.2% to 86.6% against Bybit (ETH: 68.5% to 82.4% and 65.4% to 82.7%). The bounds need no assumption about the undecided blocks, in which both premiums lie in the band (Figure 3(c,d)). By quarter the lower bound never falls below one half (lowest 50.4%, ETH against Bybit in 2025Q1); by month the order reverses for ETH in April 2025, when Hyperliquid's premium was higher in only 32% to 33% of the blocks against Binance and 29% to 34% against Bybit.

5 The premium gap

5.1 Measurement

Since all three venues apply the same band, which of them prints the floor depends only on where its averaged premium sits. We therefore compare the premiums directly, at Hyperliquid's resolution. For each hour we average the closes of Binance's and Bybit's one-minute premium-index klines over the hour Hyperliquid's record averages and define the gap $D_t = P_t^{\mathrm{HL}} - P_t^{X}$ in eight-hour-equivalent bps, X being the exchange (Appendix B). The construction reproduces the exchanges' own figures. For BTC, the close of the last minute of each hour equals the close of the exchange's own hourly premium-index kline, shipped in `data/interim`, in all 5,001 hours of the estimation window; and from 11 December 2023 to 30 September 2026, eight-hour averages with each exchange's weights reproduce the settlements that identify the premium to a median absolute error of 0.03 bps on Binance (93% within 0.1 bps; ETH 0.04 bps and 87%) and 0.003 bps on Bybit (all within 0.1 bps for both assets), once Bybit's switch from equal to rising weights on 7 March 2024 is allowed for.

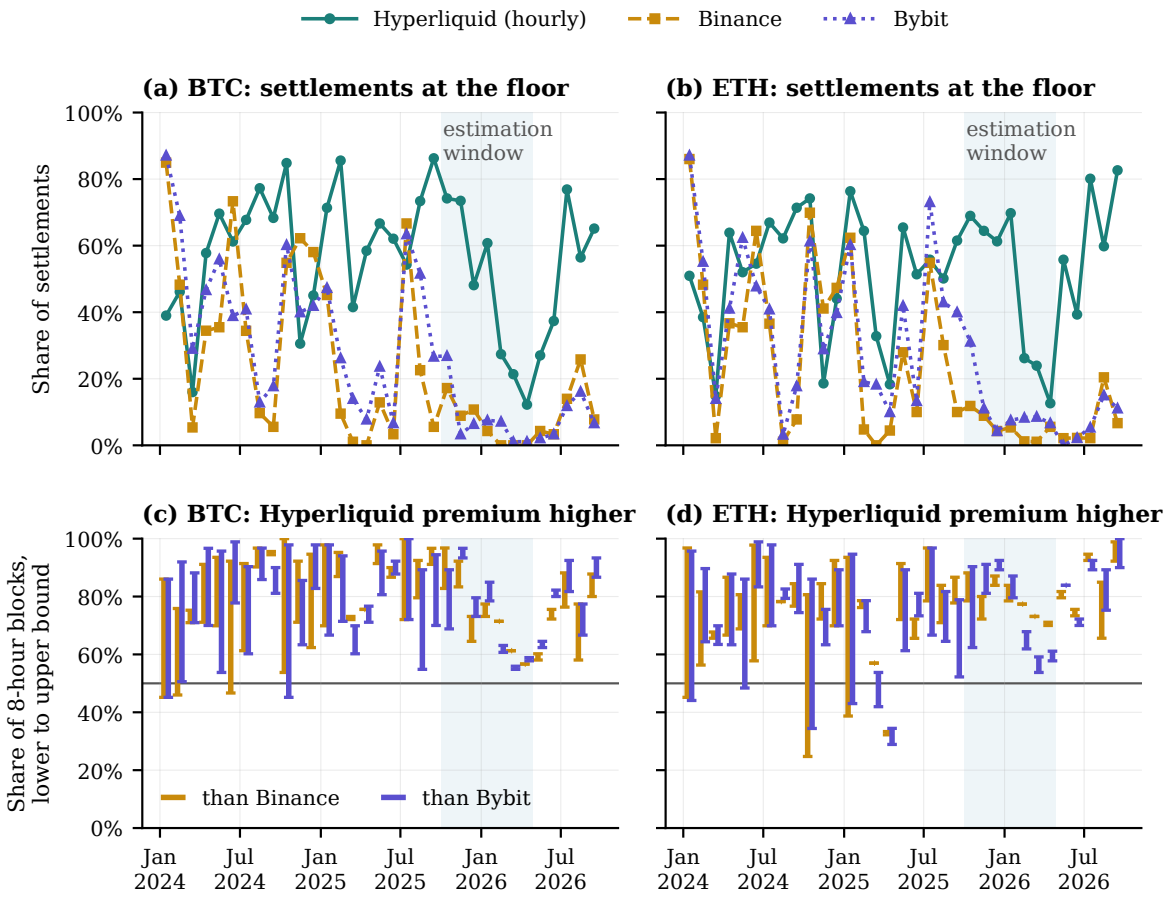


Figure 3. Censoring and the order of the premiums since December 2023, monthly. **(a, b)** Share of settlements at the floor: Hyperliquid hourly, Binance and Bybit every eight hours, BTC and ETH; the estimation window is shaded. **(c, d)** Share of eight-hour settlement blocks in which Hyperliquid's mean premium is higher than the exchange's, drawn from its lower bound (undecided blocks counted against Hyperliquid) to its upper bound (counted for it); end marks show both bounds, and a month without undecided blocks shows one mark. The partial December 2023 is left out.

5.2 Size, sign and persistence

From 11 December 2023 to 30 September 2026 Hyperliquid's BTC premium exceeded Binance's in 80% of hours and Bybit's in 81%, by medians of 2.19 and 2.20 bps; for ETH the medians are 1.69 and 1.67 bps, in 77% of hours (Table 5, Panel A; Figure 4(a,b)). The quarterly median is positive in all twelve quarters for every pair, and its 95% interval, from a moving-block bootstrap with one-week blocks, excludes zero in every full quarter. The gap is persistent in sign but not in size: for BTC its yearly median was 3.8 bps in 2024, 3.0 to 3.1 bps in 2025 and 0.6 bps in 2026 (to September); by quarter it fell to 0.3 to 0.5 bps in 2026Q1 and 2026Q2 and widened again to 1.3 to 1.5 bps in 2026Q3 (ETH 2.4, above its 2025 median). For BTC, the interval of the monthly median covers zero in April and May 2026 against both exchanges under every block length, and in a few other months that differ with the block length (with one-week blocks, five full months in all against Binance and three against Bybit). Deviations of the daily mean gap from its average halve in under three days (2.8 for BTC against Binance; about two days around each quarter's mean), though the autocorrelation at 30 days is still 0.21: the level itself moves slowly. For ETH the weekly mean gap was negative for four weeks from 31 March 2025 against both exchanges.

The gap is not a constant offset. Over the estimation window the quantiles of Hyperliquid's eight-hour BTC block means lie 0.20 bps below Binance's at the 5th percentile, 0.73 bps above at the median and 4.72 bps above at the 90th: the exchanges' premiums are packed just below the band, Hyperliquid's spread across it, and moving Hyperliquid's block means down by the difference of the two medians still leaves 34.0% of them in the band, against Binance's 5.9% at the floor. Pooled from December 2023, the gap is larger when the exchange's premium lies inside the band (BTC medians of 3.91 and 3.78 bps, against 1.69 and 1.66 below it), but four in five below-band hours fall in 2025 and 2026, when the gap was smaller, and within 2024 the ordering reverses (5.05 and 5.14 bps below the band against 3.96 and

Table 4. Funding settlements at the floor and median premiums since Hyperliquid adopted the exchanges' clamp, by asset and period. HL hourly: Hyperliquid's hourly settlements; HL 00/08/16: the same, read at the exchanges' settlement stamps. Median premium: Hyperliquid's hourly average and the exchanges' premium recovered from settlements; 'in band' marks a median that falls on censored settlements. A settlement belongs to the calendar period in which its premium accrued, and to a window by its stamp, as in Section 3: from 2 October 2025 00:00 to 28 April 2026 08:00 UTC, both included, and after it to 1 October 2026 00:00; 2023 starts on 11 December, and 2026 ends on 30 September.

	Settlements at the floor (%)				Median premium (bps)		
Period	HL hourly	HL 00/08/16	Binance	Bybit	HL hourly	Binance	Bybit
A. BTC							
11 Dec 2023 – 30 Sep 2026	55.2	54.4	23.8	27.3	−1.01	−4.42	−4.44
2023, from 11 Dec	23.7	16.4	39.3	23.0	8.84	6.51	6.77
2024	55.3	51.5	42.2	45.0	4.03	in band	in band
Jan–Jun	48.2	41.0	46.9	54.4	5.33	in band	in band
Jul–Dec	62.4	61.8	37.5	35.7	2.60	−4.09	−4.06
2025	66.1	66.9	17.2	25.5	−1.30	−4.52	−4.49
Jan–Jun	64.0	64.1	12.2	21.0	−1.33	−4.60	−4.56
Jul–Dec	68.2	69.7	22.1	29.9	−1.28	−4.46	−4.43
2026, Jan–Sep	42.9	44.6	6.7	6.3	−4.22	−4.68	−4.77
Jan–Jun	31.1	33.1	2.0	3.7	−4.56	−4.85	−4.89
Jul–Sep	66.2	67.0	15.9	11.6	−3.15	−4.39	−4.58
2 Oct 2025 – 28 Apr 2026	46.0	48.4	5.9	7.7	−4.11	−4.75	−4.82
28 Apr – 30 Sep 2026	51.7	52.7	10.9	7.9	−3.90	−4.51	−4.68
B. ETH							
11 Dec 2023 – 30 Sep 2026	52.0	51.4	23.1	28.3	−1.78	−4.42	−4.38
2023, from 11 Dec	18.0	16.4	32.8	27.9	8.86	6.85	6.78
2024	51.1	49.3	39.7	41.6	3.33	in band	in band
Jan–Jun	45.7	44.0	45.4	51.3	5.68	in band	in band
Jul–Dec	56.4	54.5	34.1	32.1	0.20	−4.06	−4.05
2025	55.9	55.7	19.4	30.7	−2.62	−4.52	−4.44
Jan–Jun	51.4	52.1	18.6	27.4	−3.35	−4.53	−4.47
Jul–Dec	60.3	59.2	20.1	33.9	−1.64	−4.52	−4.41
2026, Jan–Sep	50.3	50.9	5.3	7.2	−3.97	−4.74	−4.72
Jan–Jun	38.3	39.2	2.9	5.5	−4.36	−4.89	−4.84
Jul–Sep	74.1	73.9	9.8	10.5	−2.17	−4.54	−4.60
2 Oct 2025 – 28 Apr 2026	47.1	47.0	5.6	11.0	−4.08	−4.78	−4.79
28 Apr – 30 Sep 2026	63.0	63.2	6.6	6.6	−3.33	−4.63	−4.68

3.92 inside it). The gap does not move with realised volatility (daily rank correlations of 0.05 to 0.06 for BTC).

5.3 What the gap accounts for

Panel B of Table 5 splits the difference between Hyperliquid's share at the floor and an exchange's into three steps: *location* (Hyperliquid's premium shifted by the quarter's median gap), *averaging* (Hyperliquid's premium averaged over the exchange's eight-hour blocks with the exchange's weights) and *dispersion* (the exchange's premium in place of Hyperliquid's at the same location; Appendix B). The split depends on the order of the three steps, so we report the Shapley value with location's range over the six orders, and group quarters by where the exchange's median hourly premium $\tilde{P}_X$ lay.

In the eight quarters in which Binance's median premium lay at or below the band's lower edge, Hyperliquid's BTC funding was at the floor in 58.1% of hours against 13.2% of Binance's settlements, and location accounts for 41.6 of the 44.9 points (19.3 to 65.2 over the orders), averaging for 2.1 and dispersion for 1.2. Of these, in the five from 2024Q3 to 2025Q4 (all but 2024Q4, when Binance's median premium lay inside the band) location accounts for about all of the difference, 51.8 of 50.0 points (42.3 of 40.9 with 2024Q4), but in the three of 2026 for 24.5 of 36.2, and for less than half in 2026Q1 and 2026Q2 (Table C2), against 55% to 70% for ETH. In those two quarters the gap was 0.3 to 0.5 bps and the median premiums of all three venues lay below the band, so that the dispersion step, how much of each premium's distribution reached into the band, carried more; in 2026Q3 location again accounts for about all of the difference (51.8 of 50.2 points). Figure 4(c) shows a quarter in which location decides.

In the two quarters in which the exchanges' premiums were positive, 2023Q4 (from 11 December) and 2024Q1, the ordering reverses: Hyperliquid's BTC share is 31.7%, against 44.9% on Binance and 54.5% on Bybit, with location negative in every order (Figure 4(d)). Quarter by quarter the reversal

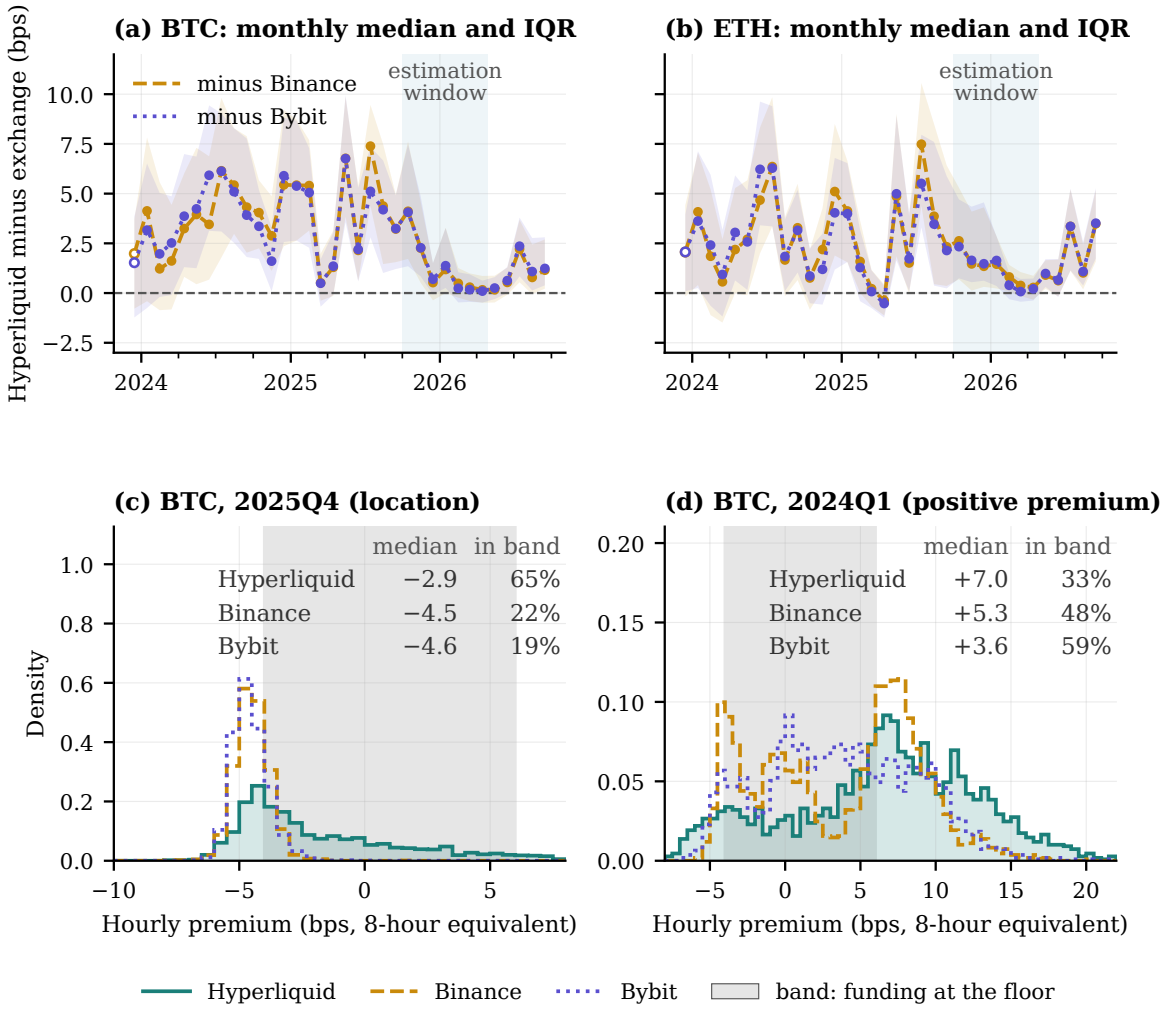


Figure 4. The premium gap, Hyperliquid minus exchange, 11 December 2023 to 30 September 2026. **(a, b)** Monthly median of the hourly gap against Binance (dashed) and Bybit (dotted) with the interquartile range shaded; the open markers are the partial December 2023, and the estimation window is shaded. **(c, d)** Hourly premiums of the three venues, BTC, in a quarter in which the exchanges' premiums lay below the band and the gap's location decides (2025Q4) and in one in which they were positive (2024Q1); grey: the band, in which funding sits at the floor. The inset tables give each venue's median premium and its share of hours in the band.

holds in seven of the eight pair-quarters, BTC against Bybit in 2023Q4 being the exception; with two quarters, one of them partial, we read it as suggestive. Where an exchange's median lay inside the band but at or below zero the shares are closer, 58.3% against 53.0% on Binance and 62.6% against 47.4% on Bybit, but location's part differs: −0.1 of 5.3 points against Binance (−13.3 to 13.5 over the orders), 14.0 of 15.2 against Bybit, whose group adds 2025Q3, when its median lay 0.09 bps inside the band. ETH shows the same reversal and closer shares, but there location exceeds the small difference, 4.0 of 2.1 and 11.6 of 4.1 points (Table C1).

5.4 Candidate sources

Why Hyperliquid's premium sits higher is not identified here. Table 1 lists the construction candidates: a median-of-medians oracle against weighted-average indices, a 20,000 USDC impact notional, a one-hour equal-weighted average against eight-hour rising weights (the averaging part of the split is small) and quanto margining; a different mix of traders is a further one. Two can be checked with public data. If traders priced Hyperliquid's book in USDC terms, its premium would fall one for one as USDC gains on USDT; the daily gap's slope on the daily mean of ln(USDC/USDT) (Binance spot, in bps) is −0.125 for BTC against Binance (HAC s.e. 0.048) and −0.07 to −0.02 for the other pairs, far from −1, and the BTC–Binance slope vanishes within quarters (−0.037, s.e. 0.038). Hyperliquid's oracle page changed from 'spot prices' to 'spot mid prices' between 6 October and 10 December 2025 (Table 1, note b); the BTC gap against Binance averaged 3.2 bps less in the 28 days from 10 December than in the 28 days to

Table 5. The premium gap and the shares at the floor. **Panel A**: median of the hourly gap $D_t = P_t^{\mathrm{HL}} - P_t^X$ (bps); 2023 runs from 11 December and 2026 to 30 September, *Estim.* is the estimation window (2 October 2025 to 28 April 2026), and *Hours* > 0 the share of hours with a positive gap over the whole span. **Panel B**: Hyperliquid's hourly share at the floor (HL), the exchange's settlement share (Exch.), their difference, and its Shapley split into location, averaging and dispersion (percentage points), with location's smallest and largest effect over the six orders; quarters are grouped by the exchange's median hourly premium $\tilde{P}_X$, their number in brackets, and the indented rows split the below-band group by period. Definitions in Appendix B; ETH in Table C1, every quarter in Table C2.

A. Hourly gap, Hyperliquid minus exchange: median (bps)

Asset	Exchange	All	2023	2024	2025	2026	Estim.	Hours > 0
BTC	Binance	2.19	1.98	3.77	3.12	0.60	0.81	80%
BTC	Bybit	2.20	1.53	3.83	2.96	0.62	0.75	81%
ETH	Binance	1.69	2.06	2.64	2.00	0.98	0.93	77%
ETH	Bybit	1.67	2.07	2.69	1.89	0.90	0.85	77%

B. Share at the floor (%) and the split of the difference (percentage points)

Quarters	HL	Exch.	Diff.	Location	[min, max]	Averaging	Dispersion
BTC, Hyperliquid against Binance							
all quarters (12)	55.2	23.8	31.5	28.7	[10.3, 48.2]	1.6	1.2
$\tilde{P}_X \leq -4$ (8)	58.1	13.2	44.9	41.6	[19.3, 65.2]	2.1	1.2
below band, 2024Q3–2025Q4 (5)	67.1	17.1	50.0	51.8	[22.9, 82.6]	1.4	−3.2
below band, 2026 (3)	42.9	6.7	36.2	24.5	[13.3, 36.0]	3.2	8.5
$-4 < \tilde{P}_X \leq 0$ (2)	58.3	53.0	5.3	−0.1	[−13.3, 13.5]	0.0	5.4
$\tilde{P}_X > 0$ (2)	31.7	44.9	−13.3	−8.8	[−11.4, −6.6]	1.0	−5.5
estimation window	46.0	5.9	40.1	30.3	[14.7, 47.0]	2.6	7.2
BTC, Hyperliquid against Bybit							
all quarters (12)	55.2	27.3	27.9	27.0	[10.4, 44.5]	1.0	−0.1
$\tilde{P}_X \leq -4$ (7)	56.2	13.9	42.3	39.5	[18.6, 61.3]	1.6	1.2
below band, 2024Q3–2025Q4 (4)	66.1	19.5	46.5	50.1	[22.0, 79.9]	1.2	−4.8
below band, 2026 (3)	42.9	6.3	36.6	25.3	[14.0, 36.5]	2.1	9.2
$-4 < \tilde{P}_X \leq 0$ (3)	62.6	47.4	15.2	14.0	[0.4, 28.8]	0.1	1.1
$\tilde{P}_X > 0$ (2)	31.7	54.5	−22.8	−12.4	[−13.8, −10.2]	−0.1	−10.4
estimation window	46.0	7.7	38.3	29.1	[14.1, 44.4]	2.1	7.1

$\tilde{P}_X$: the exchange's median hourly premium in a quarter (bps).
Shapley values; [min, max]: location's effect over the six orders.

6 October (1.3 to 3.1 bps less for the other pairs), but among all 905 placements of two 28-day windows with 64 days between them, the mean moves at least as much, up or down, in 12% (a fall at least as large in 9%; 17% to 62% for the other pairs), so the change can be neither excluded nor identified as a cause. The gap agrees in sign with the funding spread of Lau [38]: a higher premium raises funding one for one outside the band and lifts it to the floor inside it.

6 Regime dynamics

6.1 The premium

Fitted to Hyperliquid's hourly rate, a two-state Markov-switching AR(1) devotes one regime to the floor, with a variance below 10^{-31} in each of four estimations: the censoring makes the likelihood unbounded there. We therefore model Hyperliquid's BTC premium y_t in bps, with the mean-adjusted form of Hamilton [26] in which the autoregressive coefficient and the variance switch as well,

$$y_t - \mu_{S_t} \;=\; \phi_{S_t}\big(y_{t-1} - \mu_{S_{t-1}}\big) + \sigma_{S_t}\,\varepsilon_t, \qquad \varepsilon_t \overset{\text{iid}}{\sim} \mathcal{N}(0,1), \tag{2}$$

where S_t is a two-state Markov chain, estimated by maximum likelihood from 40 estimations of 40 random initialisations each (Appendix A). The within-regime half-life $\tau_k = \ln 0.5 / \ln|\phi_k|$ counts the hours in which half of a dislocation decays while the chain stays in regime k.

Within regimes the premium halves in 35.6 hours when calm and 3.2 hours under stress (Table 6), a ratio $R = \tau_{\text{calm}}/\tau_{\text{stressed}}$ of 11.2. But the stressed regime lasts 2.6 hours on average, and propagated through the fitted chain a dislocation halves after 16.3 hours when it arises under stress and 24.5 hours

Table 6. The two-state MS-AR(1) of equation (2) on the premium, estimation window, with the fitted chain and robust inference. $\hat{\mu}_k$ is the regime level, not an intercept. Half-life intervals transform $\hat{\phi}_k$'s Wald interval; an end at ∞ means that interval reaches a unit root. $\hat{\tau}_k^{\text{sw}}$ is the first hour at which the expected deviation from regime k falls to one half under the fitted chain. Side of unity: the Wald test of $\phi_{\text{calm}} = \phi_{\text{stressed}}$ at 5% (n.s.: not significant); the 95% sets are descriptive.

	Calm	Stressed
$\hat{\mu}_k$ (bps)	−2.78	−3.22
$\hat{\sigma}_k$ (bps)	0.525	2.681
$\hat{\phi}_k$	0.9807	0.8040
s.e. (Hessian; sandwich)	(0.0037; 0.0069)	(0.0462; 0.1727)
t against unity (Hessian; sandwich)	−5.28; −2.78	−4.24; −1.13
$\hat{p}_{kk}$	0.974	0.616
s.e. (Hessian; sandwich)	(0.004; 0.012)	(0.044; 0.089)
Expected duration $1/(1-\hat{p}_{kk})$ (h)	38.7	2.6
Stationary share	0.937	0.063
Half-life within the regime, $\hat{\tau}_k$ (h)	35.6	3.2
95% CI from $\hat{\phi}_k$, Hessian	[25.8, 56.8]	[2.05, 6.22]
95% CI from $\hat{\phi}_k$, sandwich	[20.7, 121.0]	[0.91, ∞)
Half-life under switching, $\hat{\tau}_k^{\text{sw}}$ (h)	25	17
interpolated	24.5	16.3
Half-life ratio	**Hessian**	**Sandwich**
$\hat{R} = \hat{\tau}_{\text{calm}}/\hat{\tau}_{\text{stressed}} = 11.19$, s.e.	4.20	12.68
Wald test of $\phi_{\text{calm}} = \phi_{\text{stressed}}$: z (p)	3.71 (<0.001)	1.01 (0.31)
side of unity	> 1	n.s.
95% CI, linear	[2.95, 19.42]	[−13.66, 36.03]
95% CI, log scale	[5.36, 23.36]	[1.21, 103.07]
95% CI, Fieller	[4.61, 23.19]	[−9.75, 70.82]
$\hat{R}^{\text{sw}} = 1.51$ (interpolated; whole hours 25/17), s.e.	0.20	0.54
side of unity	> 1	n.s.
95% CI, log scale	[1.15, 1.97]	[0.74, 3.06]
95% interval, simulated	[1.20, 2.11]	[0.00, 3.14]
$\hat{p}_{cc} + \hat{p}_{ss} - 1 = 0.590$, s.e.	0.046	0.082

$\log \mathcal{L} = -4{,}808.53$, $T_{\text{eff}} = 5{,}000$; s.e. of $\hat{R}$ from the outer product of gradients: 1.84
Largest single-hour share of the sandwich variance of $\hat{R}$: 76.7%, next largest 1.2%. It belongs to the record stamped 10 Oct 2025 23:00 UTC (premium averaged over 22:00–23:00), with $y_t = -3.13$ bps after +23.61 bps an hour earlier.

when it arises in the calm regime, a switching-adjusted ratio R^{sw} of 1.51. A single-regime AR(1) on the same premium gives a half-life of 14.5 hours; estimates elsewhere are shorter, and for other objects: a median of 3.5 hours for the basis across 41 dYdX markets [3] and about four hours for Hyperliquid's censored BTC funding rate, two to six hours across BTC, ETH and SOL [39].

The residuals are far from Gaussian, with an excess kurtosis of 78.9 (Table C3), so we add the misspecification-robust sandwich covariance [52] to the Hessian. Since, with both coefficients positive as in every fit, $R = 1$ exactly when $\phi_{\text{calm}} = \phi_{\text{stressed}}$, a Wald test of that restriction decides the side of unity for R and R^{sw} alike (Appendix A). The asymmetry is significant with the Hessian ($z = 3.71$, $p < 0.001$) but not with the sandwich ($z = 1.01$, $p = 0.31$), and 76.7% of the sandwich variance of R comes from one hour: the record stamped 10 October 2025 23:00 UTC, whose premium fell back to −3.13 bps from the series' maximum of +23.61 an hour earlier. Both records are genuine: the maximum averages the hour, 21:00 to 22:00 UTC, in which Hyperliquid's auto-deleveraging closed some $2 billion of positions [11, 13, 14], and the 23:00 record the hour after it, in a cascade whose effects on the venue's liquidity lasted weeks [40, 41]. The one-hour spike to the maximum is an additive outlier [23], the kind of observation that distorts estimates of an autoregressive coefficient [16], and a two-state model can devote its stressed regime to it.

Figures 5 and 6 measure what the hour carries. Replacing the maximum by the next largest value, 9.52 bps, lowers R to 3.04 (Hessian $p = 0.035$, sandwich $p = 0.30$); deleting it leaves 1.83, significant under neither; winsorising the upper 1%, which caps the maximum and 49 further hours, 40 of them from October 2025, at 6.75 bps, leaves 4.49, significant under both (Table C5). The hour thus sets the size of the asymmetry, and the treatment of October's upper tail decides whether the sandwich supports what remains: capping only October's 41 upper-tail hours leaves 3.93 (sandwich $p = 0.036$), while from 11 October on the winsorised series shows no asymmetry ($R = 1.11$; $p = 0.79$ and 0.87). On the hourly sub-period, which lacks October, R is 0.29, below unity with the Hessian ($p < 0.001$) but not with the sandwich ($p = 0.093$). On the five months after the window it is 0.23, below unity under

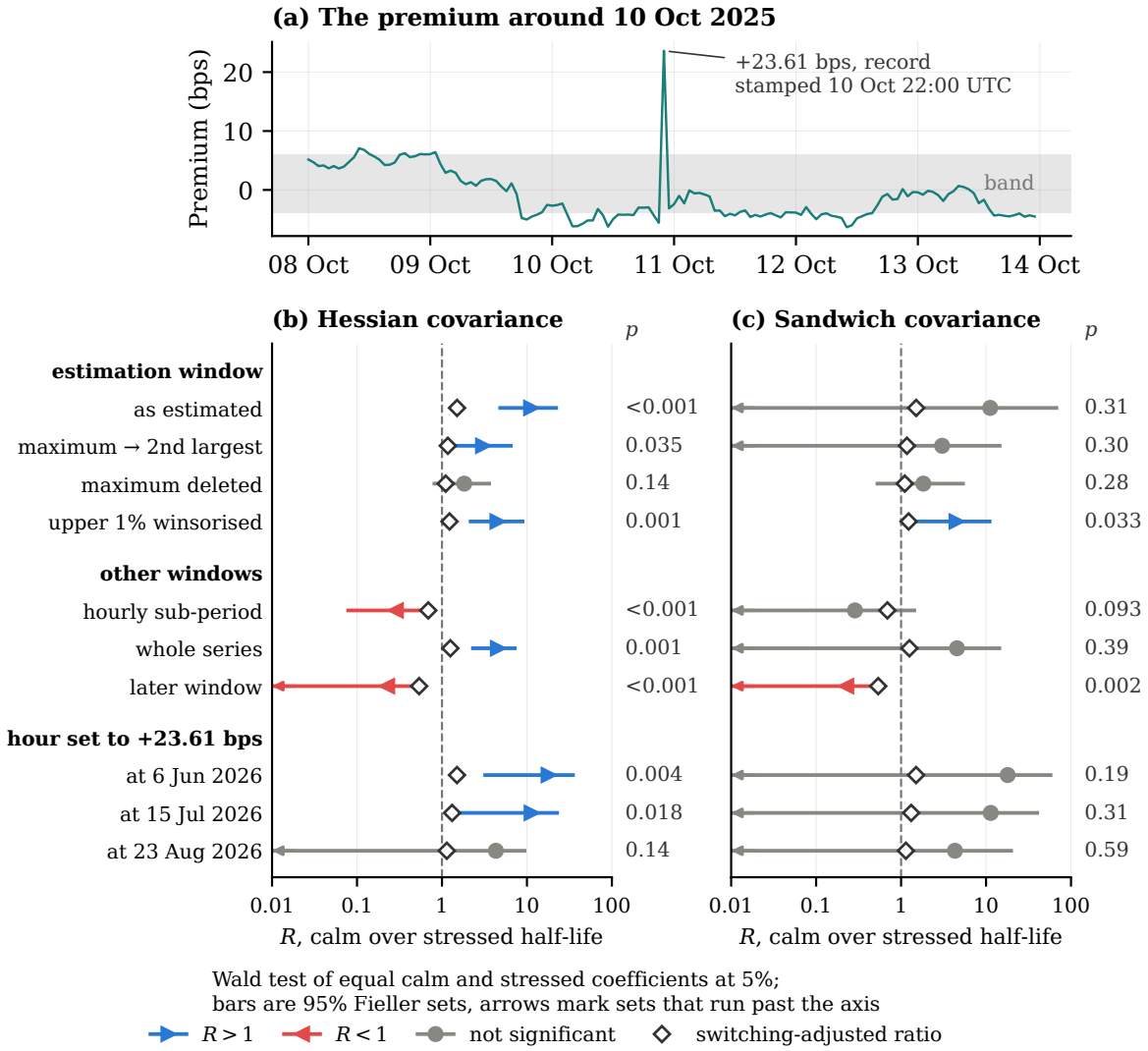


Figure 5. The hour that sets the size of the asymmetry. **(a)** Hourly premium, 8 to 13 October 2025, against the band. **(b, c)** R (filled markers) and R^{sw} (open diamonds) on the estimation window and its upper-tail variants, on the other windows, and on the later window with one hour set to the October maximum, under the Hessian (b) and the sandwich covariance (c). Whole series: the estimation and later windows together, 2 October 2025 to 30 September 2026 ($T = 8{,}737$). Colour and marker give the Wald test at 5% (blue: $R > 1$; red: $R < 1$; grey: not significant), with its p-value at the right; the bars are 95% Fieller sets, descriptive only, and arrows mark sets that run past the axis.

both covariances ($R^{\mathrm{sw}} = 0.54$); one hour of that later window set to +23.61 bps raises it to 4.3, 11.4 or 18.0, depending on where it falls. Of 14 rolling windows of 2,500 hours (Table C4), the two earliest lie above unity with the Hessian and none with the sandwich, while six lie below it under both. The second starts on 20 October, after the hour, and still lies above unity with the Hessian ($R = 2.94$, $p = 0.031$), though not with the sandwich ($p = 0.082$). The asymmetry describes the October 2025 episode and its aftermath. A size-of-prize argument, by which dislocations large enough to pay for the offsetting trade are arbitraged faster, would give its sign; Shek [49] finds the opposite ordering for extreme funding on Binance, and only several such episodes could tell the two apart.

The likelihood is also multi-modal (Figure 7). Each of our 240 estimations on the estimation window reaches one of two maxima; the inferior one, 20.14 log-likelihood points lower, attracts 20 to 30 of every 40 estimations and implies $R = 24.6$. A Markov-switching result resting on a nonlinear function of $\hat{\phi}$ therefore needs a recorded multi-start protocol, a reported terminal log-likelihood and a test rather than an interval: over the 24 fits of Tables C4 and C5 and both covariances, the log-scale delta interval for R contradicts the Wald test in 6 of 48 cases, the linear one in 8 and Fieller's set in 1.

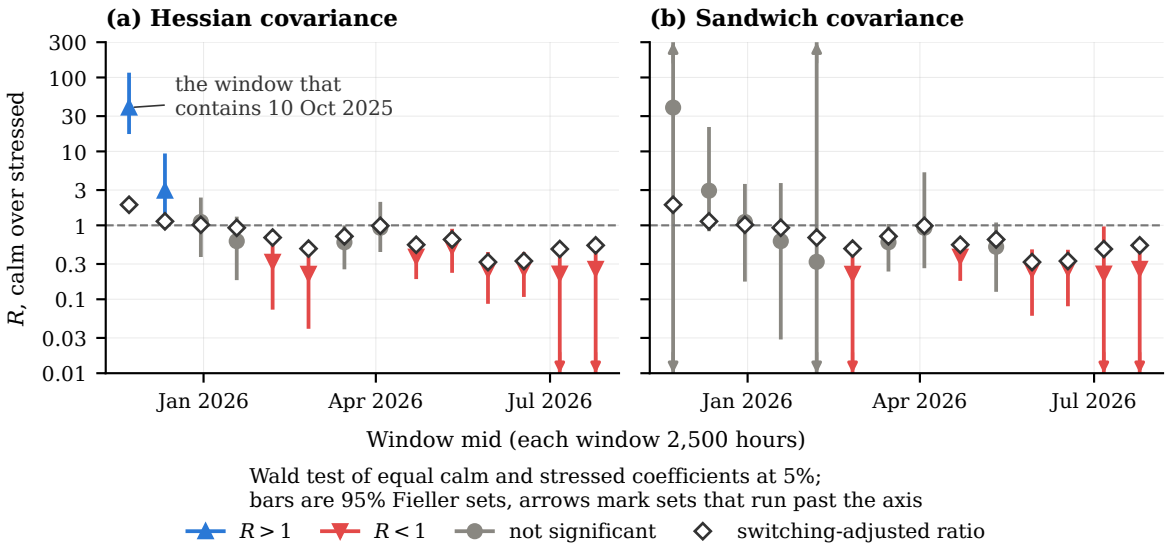


Figure 6. Rolling estimates of R (filled markers) and R^{sw} (open diamonds) at the window mid-points, windows of 2,500 hours stepped by 450, under the Hessian (a) and the sandwich covariance (b). Colours, markers and bars as in Figure 5.

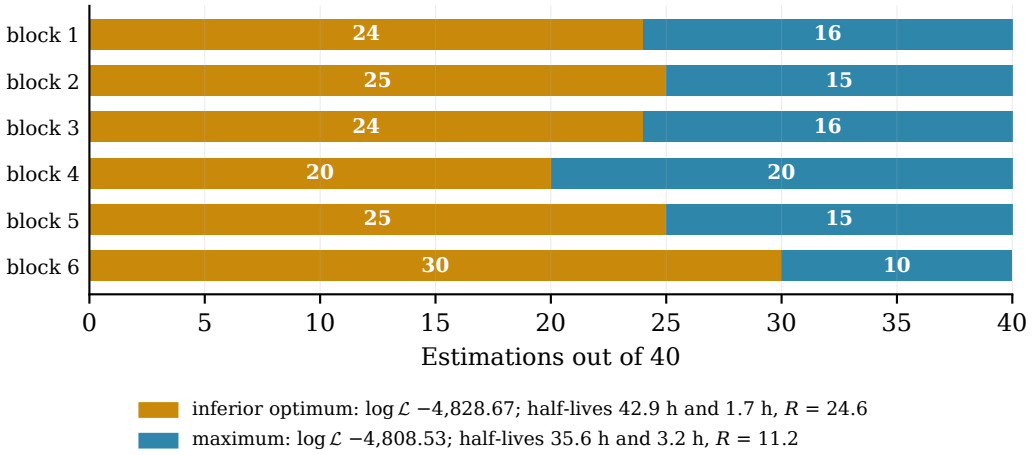


Figure 7. The two optima of the premium likelihood on the estimation window. Each bar is a block of 40 consecutive seeds, the first starting at seed 20260802, split by the optimum each estimation reaches; the legend gives what each optimum implies.

Table 7. The three-state Gaussian HMM on hourly log-returns, estimation window (T = 5,001). Annualised volatility is $\hat{\sigma}_k$ times $\sqrt{8{,}760}$; the expected duration is $1/(1-\hat{A}_{kk})$. Parenthesised figures are parametric-bootstrap standard errors from 500 replications.

State	$\hat{\mu}_k$ (bps/h)	$\hat{\sigma}_k$ (bps/h)	Annualised $\hat{\sigma}_k$	$\hat{A}_{kk}$	$\mathbb{E}[\tau_k]$ (h)
1 low	0.139	15.55	14.6%	0.9235	13.1
	(0.62)	(0.52)	(0.49)	(0.0120)	(2.08)
2 moderate	0.404	35.57	33.3%	0.8867	8.8
	(0.78)	(0.87)	(0.82)	(0.0114)	(0.90)
3 high	−5.685	94.42	88.4%	0.7307	3.7
	(3.28)	(2.97)	(2.78)	(0.0303)	(0.42)

6.2 Returns

A three-state Gaussian hidden Markov model on hourly BTC log-returns, with states ordered by volatility (Appendix A), separates the estimation window into 14.6%, 33.3% and 88.4% annualised volatility, with stationary masses of 20%, 60% and 20% and expected durations of 13.1, 8.8 and 3.7 hours (Table 7):

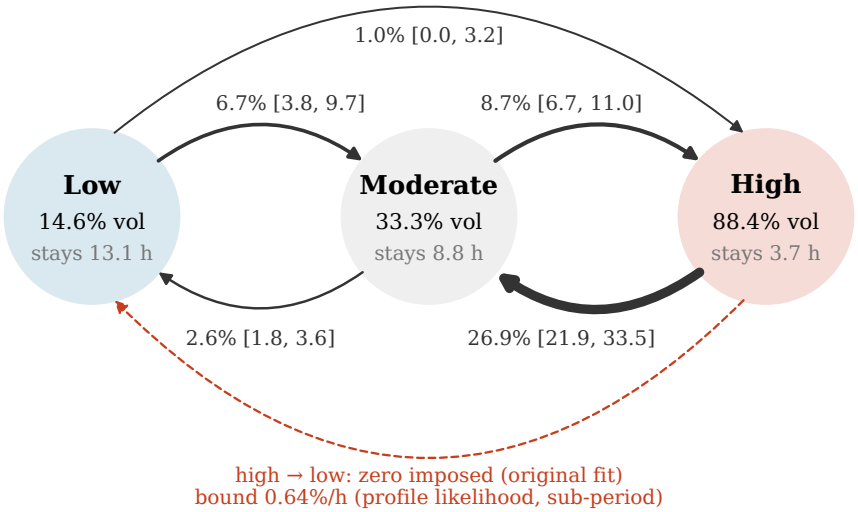


Figure 8. Transition geometry of the three-state return model on the estimation window: hourly transition probabilities with 95% parametric-bootstrap intervals, arrow width increasing with the probability; each node gives the state's annualised volatility and expected duration. The archived fit kept only the diagonal of $\hat{A}$ and the stationary distribution, so the off-diagonal cells are reconstructed with the high-to-low cell set to zero, as the original fit reported it to three decimals; that zero is an input here, and the bound on its arrow is the profile-likelihood bound of the hourly sub-period.

a classifier of hourly conditional volatility, not of market phases, in which no drift separates from zero. This fit stopped at 150 iterations and cannot be refitted, since its hourly prices before 3 December 2025 are no longer served. On the sub-period, where every fit converges, stopping at 150 iterations would change no annualised volatility by more than 0.04 percentage points. The converged fit gives 12.8%, 31.0% and 87.6%, and clipping the extreme 0.1% of returns, four hours, lowers the high state to 85.4% (Table C6). Two-hourly candles over the whole estimation window give 11.0%, 31.7% and 88.9%, with a high-to-low cell of 1.1% per two-hour bar, as expected when a bar can contain a passage through the moderate state (the reconstructed hourly chain of Figure 8 gives 0.7% over two hours); the zero of the next paragraph is an hourly statement.

Exits from the high state route through the moderate one ($\hat{A}_{32} = 0.27$; Figure 8); the direct move to the low state, $\hat{A}_{31}$, was zero to three decimals in the original fit. On the sub-period, where the fit converges, the estimate lies exactly on the boundary, and the profile likelihood bounds A_{31} below 0.64% per hour (one-sided 95%; 0.92% two-sided), a bound that is conservative within the model (1.0% rejections at a nominal 5%). A zero estimate does not show the transition to be rare: with $A_{31} = 0.3\%$ per hour in the estimation-window chain, 39% of 1,000 simulated samples still return exactly zero, and 19% at 1%. The zero is what three Gaussian states make of persistent volatility: fitted to 60 paths of the Student-t GARCH(1,1) of Table C7, a single-regime model, the three-state model returns $\hat{A}_{31} = 0$ on every one, and on 53.8% of Gaussian-GARCH paths at the best of 32 starts per path (40.4% at the eight starts). Volatility steps down through the intermediate level, consistent with volatility clustering [15], not with a barrier specific to this venue.

The number of states is descriptive. On the sub-period, with every fit converged from about 200 starts per K, BIC prefers K=4 to K=3 by 23.6 points, while K=3 gains 176.6 on K=2; under Gaussian emissions that the diagnostics reject, BIC counts the components a heavy-tailed, clustered distribution needs, not regimes. The four-state kernel has no direct exit from the top state to the bottom either, and its fourth level has no separate economic reading, the ground on which Ang and Timmermann [6] advise choosing the number of regimes; we keep K=3, as in the estimation-window fit. A Student-t GARCH(1,1) beats the three-state model on BIC with five parameters against fourteen (Table C7), so the hidden Markov model is a descriptive device. Figures C1 and C2 show the sub-period fit.

7 Implications and limitations

Headline funding is a censored measurement of the premium, and how often it is censored depends on the venue and the market phase. Funding compared across venues, whether as a gauge of positioning or of integration [53, 54], sets a series censored about half the time against series whose censored share

depends on the phase: 6% to 8% for BTC over the estimation window, about a quarter since December 2023, and in the first half of 2024 about as high as Hyperliquid's or higher (Table 4). The premiums, which the rule does not censor, are the better basis, read with their constructions in view (Table 1). Funding-arbitrage studies rightly use the cash flow [38, 51], which the censoring does not distort. For position sizing, the filtered state probabilities give one-step conditional moments that a mean-variance rule $w_t = \mathbb{E}[r_{t+1} \mid \mathcal{F}_t]/(\lambda \operatorname{Var}(r_{t+1} \mid \mathcal{F}_t))$ would turn into time-varying positions. No state's drift separates from zero (Table 7), and drifts set to zero would give no position; a common non-zero drift μ, in effect a fixed directional view, gives $w_t = \mu/(\lambda \sum_k \mathbb{P}(S_{t+1}{=}k \mid \mathcal{F}_t)\, \sigma_k^2)$, which scales that view by the inverse of the one-step conditional variance: volatility timing, not a volatility target. For execution, the stressed premium regime lasts 2.6 hours on average, and allowing for switching a dislocation halves in 16 to 25 hours whichever regime it starts in; whether a schedule could exploit that is a backtest's question we do not answer.

Every result is in-sample and descriptive: posteriors use parameters estimated on the window they describe, and we run no walk-forward, backtest or transaction-cost exercise. The censoring ordering reverses in the two quarters in which the exchanges' premiums were positive, one of them partial, and otherwise only in isolated months (Hyperliquid below both exchanges in November 2024 and, for BTC, July 2025) and against Binance over 2024Q4; these are too few to characterise when it reverses. The premium gap is a difference between venue-specific premium measures, each taken against the venue's own index or oracle with its own impact size and weights, and the candidate sources cannot be separated with public data; Bybit's index weights are snapshots, and its impact notional is not given in the pages we used. The MS-AR is misspecified: the premium index is a within-hour average, which induces a moving-average component, and the residuals are far from Gaussian; the sandwich covariance repairs its standard errors only in part, because one hour dominates it. The return model's Gaussian emission is rejected, a Student-t GARCH fits better on BIC, and the estimation-window fit stopped at 150 iterations and cannot be re-estimated; its sub-period counterpart is converged. The documentation behind Table 1 is as retrieved on 8 October 2026 and need not describe the whole sample.

8 Conclusion

One funding rule, common to Hyperliquid, Binance and Bybit, pins the rate to the interest rate whenever the averaged premium lies in a band 10 bps wide. Measured exactly, Hyperliquid's BTC funding sat there in 46% of hours over the estimation window and 55% since December 2023, against 6% to 8% and 24% to 27% of the exchanges' settlements, and ETH is similar. The difference is a difference in premiums: Hyperliquid's sat a median 2.2 bps above the exchanges' for BTC, persistent in sign, smallest in the first half of 2026 and wider again in its third quarter, and in the two quarters in which the exchanges' premiums were positive, one of them partial, the ordering reversed, which we read as suggestive. Cross-venue work that reads funding as positioning or integration should compare premiums, with their constructions in view; why Hyperliquid's premium sits higher is the question the measurement opens.

The regime models describe one asset, one venue and one episode. The premium's faster reversion under stress belongs to the October 2025 episode and its aftermath: on the estimation window it is significant with the Hessian but not with the sandwich covariance, and one hour of the 10 October crash sets most of its size, the ratio falling from 11.2 to 1.8 without that hour and to 3.0 with the hour set to the next largest value. With the premium's upper 1% winsorised it is significant under both covariances (sandwich $p = 0.033$), and the first rolling window to start after the crash, on 20 October, still finds it with the Hessian ($p = 0.031$); later data reverse it, and its likelihood has a frequent optimum that is not its maximum. The return model's three volatility states and its high-to-low transition at zero discretise clustered volatility in a way a single-regime GARCH reproduces. Natural extensions are a Student-t or generalised-hyperbolic emission [22], within-state volatility dynamics [25], time-varying transition probabilities [21], more venues, assets and phases, and a carry backtest once the asymmetry has been seen across several episodes.

Data and Code Availability

All data come from unauthenticated public endpoints. Hyperliquid: `fundingHistory` (the funding rate and premium of each settlement) and `candleSnapshot` (candles of traded prices). Binance: `fapi/v1/fundingRate`, `fapi/v1/premiumIndexKlines`, `fapi/v1/klines` and the spot `api/v3/klines` for USDCUSDT. Bybit: `v5/market/funding/history` and `v5/market/premium-index-price-kline`. The original hourly extract of the estimation window was retrieved on 28 April 2026 and survives

only through its archived summary (fitted parameters and price range); the shipped hourly panel, from 3 December 2025, and its continuation to 27 August 2026 (Figure 1) were retrieved from `candleSnapshot` after that date, by 2 August and 3 September 2026 respectively. The premium series of the estimation and later windows was re-retrieved from `fundingHistory` on 19 August 2026 and extended on 6 October 2026, when the Binance and Bybit series of those windows were retrieved as well; the two-hourly candles were retrieved in August and October 2026, and the long histories (`data/longhistory`) and the premium-gap series (`data/premium_gap`: the hourly aggregates of the one-minute premium-index klines, settlements and prices) on 8 October 2026. No proprietary or licensed data are used.

Every funding, premium, censoring, premium-gap and MS-AR result rebuilds from the shipped series, which can themselves be re-fetched. The hourly prices before 3 December 2025 cannot: the estimation-window return fit (Table 7, Figure 8 and the volatilities quoted from it) rests on the original extract through its archived parameters, and the off-diagonal cells of its transition matrix are reconstructed. Code, data, the fitted parameters behind every table, figure and quoted number, and the seeds and multi-start protocols are in the replication repository (version 1.3.0),

`https://github.com/gelatotrade/HMM-Decentralised-Perpetual-Futures/tree/v1.3.0`.

Use of Generative Tools

AI assistants (Claude, Anthropic) supported this work: in searching the literature, in writing and testing code, in cross-checking results and in drafting and editing the text. The research question, the design of the study and the interpretation of the results are the author's; the author directed the work, reviewed its results and takes full responsibility for the content of this paper.

Competing Interests

The author trades on Hyperliquid with personal funds and holds Hyperliquid-related assets; the author does not trade on Binance or Bybit and has no equity, advisory or employment relationship with any of the three venues. The author contributes, unpaid, to the Hyperliquid Research Collective (HRC), which republishes this research free of charge alongside this posting. HRC was initiated by GLC Research and Four Pillars with support from HypurrCollective, Hyperion DeFi, Hyperliquid Strategies and B-Harvest (its website now lists Altus in place of B-Harvest); its launch announcement states the aim of lowering perceived market risk and improving the valuation multiples of Hyperliquid's token, and two of its supporters are listed companies that hold that token. HRC describes itself as independent research, not affiliated with Hyperliquid. Neither HRC nor its supporters influenced the choice of topic, reviewed the analysis or approved this paper, and the author received no compensation connected with this work.

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A Estimation details

The return model. With $S_t \in \{1, \dots, K\}$ the latent state and Gaussian emissions, the filtered posterior $\mathbb{P}(S_t = k \mid \mathcal{F}_t)$ uses only the past; the smoothed posterior conditions on the whole sample and is an in-sample diagnostic only. Baum–Welch runs in log space. Every random start (eight per fit; on the data seeded $42 + K$, that is 45 for $K = 3$ and 44 for $K = 2$, and on simulated paths one seed per path) runs until the log-likelihood gains less than 10^{-6}; the first start with the largest final log-likelihood is continued until it changes by less than 10^{-9} and no parameter moves by more than 10^{-10}. For the comparison of $K = 2$ to 5 each fit starts from about 200 points: 40 random starts drawn like the eight (seed $45+K$), 160 with state volatilities spread log-uniformly over 0.15 to 3.5 times the sample standard deviation (seed $20261009+K$), and splits of each state of the best fit with one state fewer. The archived estimation-window fit used the eight starts but stopped at 150 iterations. Standard errors come from a parametric bootstrap of 500 series simulated from the fitted chain. States are ordered by $\hat{\sigma}_k$: ordered by $\hat{\mu}_k$ [50], the labels would switch in 40.2% of bootstrap replications, because two of the three means lie within 0.27 bps per hour of each other, and the 95% interval for the state with $\hat{\sigma} = 15.6$ bps per hour would be $[14.6, 90.7]$ instead of $[14.5, 16.5]$.

The premium model. In (2) μ_k is the regime level; the fixed-coefficient form would carry the intercept $\mu_k(1 - \phi_k)$, which is badly behaved with $\hat{\phi}_k$ near unity. We estimate with `statsmodels`, which combines an EM initialisation, a random start search and quasi-Newton optimisation with the Hamilton filter and the smoother of Kim [35], from 40 estimations (seeds 20260802 to 20260841) of 40 random initialisations each, and report the highest terminal log-likelihood. An estimation whose start search fails numerically is dropped, and a start whose log-likelihood is not finite, whose smallest regime variance falls below 10^{-6} times the sample variance or whose Hessian is singular cannot be the reported optimum.

Half-lives under switching. With P the transition matrix and $\Phi = \mathrm{diag}(\phi_1, \phi_2)$, the expected deviation $x_t = y_t - \mu_{S_t}$ after a unit dislocation in regime k is

$$m_k(h) = \mathbb{E}[x_{t+h} \mid x_t = 1, S_t = k] = e_k^\top (P\Phi)^h \mathbf{1}, \tag{A1}$$

and τ_k^{sw} is the first hour with $m_k(h) \le 1/2$, interpolated log-linearly between whole hours. The ratio R^{sw} of the calm to the stressed τ_k^{sw} exceeds unity exactly when $\phi_{\mathrm{calm}} > \phi_{\mathrm{stressed}}$, provided both ϕ_k are positive and $p_{cc} + p_{ss} - 1$, the second eigenvalue of P, is positive; it therefore takes the Wald verdict whenever both coefficients are positive and this eigenvalue is significantly positive, as they are in every fit we report. Its simulated interval draws 50,000 parameter vectors from their asymptotic normal distribution, the transition probabilities on the logit scale.

Covariances and intervals. The Hessian covariance is $-H^{-1}$, the outer product of gradients is $(S^\top S)^{-1}$, with S the matrix of per-hour scores, and the sandwich is $H^{-1} S^\top S H^{-1}$ [52]. The side of unity of R is the two-sided Wald test of $\phi_{\mathrm{calm}} = \phi_{\mathrm{stressed}}$ at 5%. The linear and log-scale delta intervals and Fieller's set [20] are descriptive; fed with delta-method variances of $\ln|\phi_k|$, Fieller's set is first-order here too, and it can be unbounded.

B The premium gap: construction and validation

Alignment and settlements. Hyperliquid stamps each funding record at the end of the hour whose premium it averages, so the record stamped t covers $[t-1\text{h}, t)$; for the same hour we average the closes of the exchange's one-minute premium-index klines that open in $[t-1\text{h}, t)$ and keep an hour only when all 60 minutes are present. An hour enters D_t only when both venues have it: one Binance hour lacks minutes, which leaves 24,576 hours against Binance and 24,577 against Bybit. An exchange settlement at s averages its premium over $(s-8\text{h}, s]$: Binance weights its five-second samples $1, \ldots, 5{,}760$, which we rebuild from minute closes with linear weights, and Bybit weights its minute samples $1, \ldots, 480$ from the settlement of 7 March 2024 16:00 UTC and equally before.

Decomposition. For a group of quarters let G be each quarter's median hourly gap. Three steps take Hyperliquid's share of hours in the band to the exchange's share of settlements at the floor: *location* shifts the premium by $-G$; *averaging* replaces hours by averages over the exchange's eight-hour blocks, weighted as the exchange weights them; *dispersion* replaces Hyperliquid's premium by the exchange's at the same location, shifted by $+G$ while the location is still Hyperliquid's. Each step is taken (1) or not (0), and each of the eight combinations (location, averaging, dispersion) gives a share in the band, lettered as in the replication package: (a) $(0,0,0)$, Hyperliquid's hours; (b) $(1,0,0)$, the same hours shifted by $-G$; (c) $(0,1,0)$, Hyperliquid's eight-hour averages; (d) $(1,1,1)$, the exchange's settlements at the floor; (e) $(1,1,0)$, the averages of (c) shifted by $-G$; (f) $(0,0,1)$, the exchange's hours shifted by $+G$; (g) $(1,0,1)$, the exchange's hours; and (h) $(0,1,1)$, the exchange's eight-hour averages, rebuilt from its minute klines, shifted by $+G$. Location thus moves a→b, c→e, f→g and h→d; averaging a→c, b→e, f→h and g→d; dispersion a→f, b→g, c→h and e→d. In a given order of the three steps, a step's effect is the fall in the share when it is taken; its Shapley value averages that effect over the six orders, and the three values add up to (a) minus (d). Tables 5 and C1 also give location's smallest and largest effect over the orders, and Table C2 dispersion's as well. Only (d) uses the settlements themselves; their small difference from the rebuilt averages enters each step's Shapley value equally.

Intervals. The 95% intervals of the monthly and quarterly medians of D_t are percentile intervals from a moving-block bootstrap with 168-hour blocks and 2,000 replicates; 24- and 72-hour blocks, run from the same seed, serve as a sensitivity check. The partial quarter 2023Q4 and month December 2023 are reported apart.

C Additional tables and figures

Table C1. Panel B of Table 5 for ETH: Hyperliquid's hourly share at the floor (HL), the exchange's settlement share (Exch.), their difference, and its Shapley split (percentage points), by group of quarters.

B. Share at the floor (%) and the split of the difference (percentage points)							
Quarters	HL	Exch.	Diff.	Location	[min, max]	Averaging	Dispersion
ETH, Hyperliquid against Binance							
all quarters (12)	52.0	23.1	28.8	29.3	[9.8, 50.9]	1.8	−2.3
$\tilde{P}_X \le -4$ (8)	55.2	13.6	41.6	41.4	[17.3, 68.0]	2.1	−1.8
below band, 2024Q3–2025Q4 (5)	58.1	18.5	39.6	45.2	[15.6, 78.2]	1.7	−7.4
below band, 2026 (3)	50.3	5.3	45.1	34.9	[20.0, 51.0]	2.7	7.5
$-4 < \tilde{P}_X \le 0$ (2)	51.3	49.2	2.1	4.0	[−7.1, 15.7]	1.0	−2.9
$\tilde{P}_X > 0$ (2)	31.7	43.1	−11.5	−7.8	[−12.6, −3.3]	1.3	−4.9
estimation window	47.1	5.6	41.5	36.5	[17.8, 56.9]	2.9	2.1
ETH, Hyperliquid against Bybit							
all quarters (12)	52.0	28.3	23.7	25.1	[8.9, 43.2]	1.3	−2.7
$\tilde{P}_X \le -4$ (7)	55.1	16.1	39.0	37.2	[17.3, 59.6]	1.8	0.0
below band, 2024Q3–2025Q4 (4)	58.7	22.7	36.0	41.4	[16.5, 69.7]	1.2	−6.6
below band, 2026 (3)	50.3	7.2	43.1	31.7	[18.1, 46.0]	2.6	8.8
$-4 < \tilde{P}_X \le 0$ (3)	52.8	48.7	4.1	11.6	[−3.3, 26.2]	0.4	−7.9
$\tilde{P}_X > 0$ (2)	31.7	47.6	−15.9	−10.9	[−13.2, −8.7]	0.4	−5.4
estimation window	47.1	11.0	36.1	31.0	[16.2, 47.8]	2.3	2.7

$\tilde{P}_X$: the exchange's median hourly premium in a quarter (bps).
Shapley values; [min, max]: location's effect over the six orders.

Table C2. The split of Table 5 by quarter, for both assets and exchanges, with the smallest and largest effect of location and of dispersion over the six orders of the three steps.

By quarter: share at the floor (%) and the split of the difference (percentage points)									
Quarter	$\tilde{P}_X$	HL	Exch.	Diff.	Location	[min, max]	Averaging	Dispersion	[min, max]
BTC, Hyperliquid against Binance									
2023Q4*	6.71	23.7	39.3	−15.7	−22.7	[−32.8, −11.5]	−1.1	8.1	[−1.6, 19.7]
2024Q1	5.27	33.4	46.2	−12.7	−5.7	[−10.6, −0.7]	1.4	−8.5	[−13.2, −3.9]
2024Q2	−2.99	63.0	47.6	15.3	5.1	[−8.2, 19.8]	0.5	9.8	[−4.7, 25.6]
2024Q3	−4.65	71.1	16.7	54.5	54.3	[26.1, 83.3]	0.5	−0.3	[−28.9, 29.0]
2024Q4	−2.26	53.7	58.3	−4.6	−5.2	[−18.5, 7.2]	−0.4	1.0	[−11.2, 14.5]
2025Q1	−4.45	65.5	18.9	46.6	50.0	[21.0, 80.7]	2.3	−5.7	[−34.2, 24.4]
2025Q2	−4.64	62.5	5.5	57.0	56.5	[20.1, 94.5]	0.9	−0.5	[−37.5, 38.1]
2025Q3	−4.23	71.2	31.9	39.3	45.1	[24.1, 67.4]	0.6	−6.5	[−28.1, 16.3]
2025Q4	−4.51	65.2	12.3	52.9	53.1	[22.8, 87.3]	2.8	−3.1	[−34.4, 30.1]
2026Q1	−4.84	36.8	1.5	35.3	15.4	[9.8, 22.3]	2.8	17.1	[7.8, 25.2]
2026Q2	−4.86	25.5	2.6	23.0	6.1	[4.0, 9.1]	3.5	13.4	[9.8, 17.6]
2026Q3	−4.37	66.2	15.9	50.2	51.8	[23.5, 83.7]	3.3	−4.8	[−34.1, 25.7]
BTC, Hyperliquid against Bybit									
2023Q4*	6.96	23.7	23.0	0.7	−16.2	[−22.4, −9.8]	3.4	13.6	[6.0, 21.3]
2024Q1	3.59	33.4	61.5	−28.1	−11.5	[−12.5, −9.8]	−0.9	−15.7	[−16.8, −13.7]
2024Q2	−3.62	63.0	47.3	15.7	10.1	[−3.7, 23.8]	−0.5	6.2	[−7.6, 21.6]
2024Q3	−4.47	71.1	23.9	47.2	50.5	[26.1, 76.1]	0.7	−3.9	[−28.5, 21.7]
2024Q4	−0.51	53.7	47.5	6.2	−0.7	[−16.4, 15.9]	0.5	6.5	[−9.1, 22.8]
2025Q1	−4.35	65.5	29.3	36.2	43.7	[19.6, 69.3]	1.6	−9.1	[−32.2, 15.6]
2025Q2	−4.65	62.5	12.8	49.6	52.9	[19.6, 87.2]	0.5	−3.8	[−37.4, 30.8]
2025Q3	−3.91	71.2	47.5	23.7	32.7	[19.7, 46.7]	0.4	−9.4	[−23.1, 5.4]
2025Q4	−4.63	65.2	12.3	52.9	53.2	[22.6, 87.0]	2.0	−2.4	[−34.1, 30.1]
2026Q1	−4.85	36.8	5.2	31.6	11.8	[8.8, 17.1]	2.5	17.3	[10.8, 21.5]
2026Q2	−4.89	25.5	2.2	23.4	7.4	[7.0, 8.0]	2.2	13.7	[12.4, 15.0]
2026Q3	−4.54	66.2	11.6	54.6	56.3	[26.1, 88.4]	1.6	−3.3	[−34.4, 27.2]
ETH, Hyperliquid against Binance									
2023Q4*	7.00	18.0	32.8	−14.7	−23.5	[−26.2, −19.7]	4.6	4.2	[0.0, 7.5]
2024Q1	5.20	34.7	45.4	−10.8	−4.3	[−10.6, 1.8]	0.6	−7.0	[−12.8, −0.9]
2024Q2	−3.26	56.8	45.4	11.4	0.9	[−15.0, 16.8]	0.7	9.8	[−5.6, 27.1]
2024Q3	−4.51	66.8	15.2	51.6	53.6	[24.7, 84.8]	1.0	−3.0	[−33.1, 29.3]
2024Q4	−1.82	45.9	52.9	−7.0	7.2	[0.7, 14.5]	1.3	−15.5	[−22.1, −8.3]
2025Q1	−4.44	57.6	23.0	34.7	42.7	[14.8, 74.8]	0.9	−8.9	[−39.3, 20.7]
2025Q2	−4.55	45.3	14.3	31.0	36.5	[5.5, 72.5]	1.9	−7.5	[−41.8, 24.5]
2025Q3	−4.32	55.8	31.9	23.9	38.2	[10.4, 67.4]	1.7	−16.0	[−43.8, 12.7]
2025Q4	−4.55	64.9	8.3	56.6	55.0	[22.6, 91.3]	3.0	−1.5	[−34.4, 32.6]
2026Q1	−4.91	40.4	2.6	37.8	26.4	[17.2, 36.3]	3.1	8.3	[−2.3, 19.6]
2026Q2	−4.87	36.1	3.3	32.8	19.5	[12.5, 26.4]	3.1	10.3	[1.6, 19.0]
2026Q3	−4.50	74.1	9.8	64.3	58.4	[29.6, 90.2]	1.9	3.9	[−25.7, 36.6]
ETH, Hyperliquid against Bybit									
2023Q4*	6.82	18.0	27.9	−9.8	−24.3	[−28.4, −21.3]	3.2	11.3	[6.6, 14.8]
2024Q1	4.29	34.7	52.0	−17.4	−7.9	[−11.3, −4.8]	−0.3	−9.1	[−12.7, −5.9]
2024Q2	−3.22	56.8	50.5	6.2	1.6	[−16.5, 16.5]	−0.7	5.4	[−10.6, 23.4]
2024Q3	−4.58	66.8	20.7	46.1	50.4	[23.4, 79.3]	0.8	−5.0	[−33.1, 25.0]
2024Q4	−0.56	45.9	43.5	2.4	9.9	[0.7, 19.6]	1.3	−8.8	[−17.8, −0.7]
2025Q1	−4.27	57.6	33.0	24.7	32.0	[13.7, 53.7]	1.1	−8.4	[−28.1, 11.9]
2025Q2	−4.51	45.3	22.0	23.3	32.3	[5.6, 63.7]	1.7	−10.7	[−40.7, 16.8]
2025Q3	−3.81	55.8	52.2	3.6	23.2	[4.0, 42.4]	0.5	−20.2	[−39.1, −0.7]
2025Q4	−4.67	64.9	15.6	49.3	50.4	[23.1, 81.5]	1.4	−2.5	[−31.9, 24.6]
2026Q1	−4.79	40.4	8.1	32.3	17.7	[12.6, 23.3]	3.4	11.2	[4.2, 19.3]
2026Q2	−4.86	36.1	2.9	33.2	18.7	[11.7, 25.7]	3.4	11.1	[2.0, 20.1]
2026Q3	−4.57	74.1	10.5	63.6	58.3	[29.3, 89.5]	1.1	4.1	[−25.8, 36.2]

* From 11 Dec 2023. $\tilde{P}_X$: the exchange's median hourly premium in the quarter (bps).
Shapley values; [min, max]: the effect of the step to its left over the six orders of the three steps.

Table C3. Residual diagnostics. Raw-return and HMM rows use the hourly sub-period ($T = 3{,}507$); MS-AR rows use the estimation window ($T = 5{,}001$, leaving 4,999 residuals). For the HMM rows z_t is standardised by the mixture mean and the full, within- plus between-state, variance; the headline rows weight the states by the one-step-ahead filtered prediction $\mathbb{P}(S_t \mid \mathcal{F}_{t-1})$, the smoothed rows by the contemporaneous smoothed posterior $\mathbb{P}(S_t \mid \mathcal{F}_T)$, which conditions on r_t itself and is shown for contrast only. Moment-standardised mixture residuals are not $\mathcal{N}(0,1)$ even when the model is correct, so their Jarque–Bera p-values are not valid tests; probability-integral-transform residuals [17, 47] would be.

Panel A: Ljung–Box $Q(\ell)$, p-value in brackets				
Series	ℓ	$Q(z)$	$Q(z^2)$	$Q(\lvert z\rvert)$
Raw log-returns	10	6.7 [0.76]	470.7 [<0.0001]	976.0 [<0.0001]
	24	17.0 [0.85]	671.2 [<0.0001]	1,718.1 [<0.0001]
HMM K=3, filtered	10	7.9 [0.64]	2.4 [0.99]	20.6 [0.024]
	24	18.2 [0.79]	5.7 [1.00]	96.0 [<0.0001]
HMM K=3, smoothed	10	14.2 [0.16]	53.4 [<0.0001]	32.8 [0.0003]
	24	28.2 [0.25]	92.5 [<0.0001]	78.4 [<0.0001]
MS-AR(1), estimation window	1	6.7 [0.0096]	48.9 [<0.0001]	36.8 [<0.0001]
	10	42.6 [<0.0001]	51.5 [<0.0001]	90.3 [<0.0001]

Panel B: heteroskedasticity and normality				
Series	ARCH-LM(12)	Jarque–Bera	Skew	Exc. kurt.
Raw log-returns	270.3 [<0.0001]	7,081 [<0.0001]	−0.26	6.94
HMM K=3, filtered	2.4 [0.9985]	31,194 [<0.0001]	−1.02	14.47
HMM K=3, smoothed	46.2 [<0.0001]	13 [0.0016]	−0.06	0.27
MS-AR(1), estimation window	51.4 [<0.0001]	1,304,072 [<0.0001]	2.80	78.93

Table C4. Rolling MS-AR(1) estimates of $R = \hat{\tau}_{\text{calm}}/\hat{\tau}_{\text{stressed}}$ on the premium series, windows of 2,500 hours stepped by 450, 40 estimations each, at the window mid-points; estimations that fail numerically (0 to 24 of the 40 per window) are dropped, and the reported optimum is reached by 3 to 40 of the rest (3 of 38 in the window centred on 6 February 2026, whose most frequent optimum lies 0.13 log-likelihood points lower); in the fit of Table C5 with one hour set to the maximum at 15 July 2026, 18 of the 40 fail numerically, 18 more end at the variance floor or with a singular Hessian, and the reported optimum is reached by 2 of the 4 admissible ones. p: Wald test of $\phi_{\text{calm}} = \phi_{\text{stressed}}$ under the Hessian and the sandwich covariance; Side: its verdict at 5% (n.s.: not significant). $\hat{R}^{\text{sw}}$: the switching-adjusted ratio, with a 95% interval simulated from the sandwich covariance; an end at 0.00 or ∞ comes from draws whose expected path does not halve within 10,000 hours.

		Wald test of $\phi_{\text{calm}} = \phi_{\text{stressed}}$				Under switching	
		Hessian		Sandwich			simulated, sandwich
Window mid	$\hat{R}$	p	Side	p	Side	$\hat{R}^{\text{sw}}$	95% interval
23 Nov 2025	39.17	<0.001	> 1	0.23	n.s.	1.90	[0.00, 5.60]
11 Dec 2025	2.94	0.031	> 1	0.082	n.s.	1.13	[0.98, 1.33]
30 Dec 2025	1.11	0.80	n.s.	0.86	n.s.	1.02	[0.81, 1.22]
18 Jan 2026	0.61	0.21	n.s.	0.49	n.s.	0.93	[0.53, 1.08]
6 Feb 2026	0.32	0.018	< 1	0.49	n.s.	0.68	[0.03, ∞)
24 Feb 2026	0.22	<0.001	< 1	0.004	< 1	0.48	[0.25, 0.82]
15 Mar 2026	0.59	0.051	n.s.	0.073	n.s.	0.71	[0.47, 1.03]
3 Apr 2026	0.93	0.84	n.s.	0.90	n.s.	0.99	[0.65, 1.15]
22 Apr 2026	0.38	<0.001	< 1	<0.001	< 1	0.55	[0.38, 0.79]
10 May 2026	0.51	0.019	< 1	0.089	n.s.	0.64	[0.46, 1.09]
29 May 2026	0.25	<0.001	< 1	<0.001	< 1	0.32	[0.16, 0.53]
17 Jun 2026	0.26	<0.001	< 1	<0.001	< 1	0.33	[0.18, 0.55]
6 Jul 2026	0.22	<0.001	< 1	0.043	< 1	0.48	[0.21, 0.97]
24 Jul 2026	0.26	<0.001	< 1	0.002	< 1	0.54	[0.30, 0.83]

Table C5. The ratio R under changes to the premium's upper tail on the estimation window, on the other windows, and on the later window with one hour set to the October maximum, +23.61 bps, at each quartile; columns as in Table C4. *Max.* → *2nd largest*: the maximum replaced by the next largest value, 9.52 bps. *Whole series*: the estimation and later windows together, 2 October 2025 to 30 September 2026.

			Wald test of $\phi_{\text{calm}} = \phi_{\text{stressed}}$				Under switching	
			Hessian		Sandwich			simulated, sandwich
Series	T	$\hat{R}$	p	Side	p	Side	$\hat{R}^{\text{sw}}$	95% interval
Estimation window, upper tail changed								
Unchanged	5,001	11.19	<0.001	> 1	0.31	n.s.	1.51	[0.00, 3.14]
Max. → 2nd largest	5,001	3.04	0.035	> 1	0.30	n.s.	1.17	[0.81, 1.51]
Max. deleted	5,000	1.83	0.14	n.s.	0.28	n.s.	1.11	[0.92, 1.43]
Upper 1% winsorised	5,001	4.49	0.001	> 1	0.033	> 1	1.23	[1.02, 1.52]
Other windows								
Hourly sub-period	3,507	0.29	<0.001	< 1	0.093	n.s.	0.69	[0.24, 1.01]
Whole series	8,737	4.55	0.001	> 1	0.39	n.s.	1.25	[0.55, 1.74]
Later window	3,736	0.23	<0.001	< 1	0.002	< 1	0.54	[0.29, 0.84]
Later window, one hour set to the October maximum								
At 6 Jun 2026	3,736	17.98	0.004	> 1	0.19	n.s.	1.50	[0.80, 6.53]
At 15 Jul 2026	3,736	11.36	0.018	> 1	0.31	n.s.	1.32	[0.71, 3.34]
At 23 Aug 2026	3,736	4.31	0.14	n.s.	0.59	n.s.	1.14	[0.52, 2.48]

Table C6. The three-state volatilities with the most extreme 0.1% of hourly returns (four) winsorised, hourly sub-period; column k is State k of Table 7. The high-to-low transition stays at zero in both fits.

Annualised $\hat{\sigma}$	1 low	2 moderate	3 high
Raw ($T = 3{,}507$)	12.8%	31.0%	87.6%
Winsorised (extreme 0.1%)	12.6%	30.4%	85.4%

Table C7. Out-of-family comparison on the hourly sub-period ($T = 3{,}507$), all rows on log-returns in decimal units (in bps each $\log \mathcal{L}$ shifts by $-T \ln 10^4$, leaving differences unchanged); AIC $= 2k - 2\log \mathcal{L}$ and BIC $= k \log T - 2 \log \mathcal{L}$, lower is better. The Student-t GARCH, best on BIC, is near-integrated ($\hat{\alpha} + \hat{\beta} = 0.99998$ against an imposed bound of 0.99999) with $\hat{\nu} = 3.35$.

Model	$\log \mathcal{L}$	# params	AIC	BIC
GARCH(1,1)-Student-t	14,312.1	5	−28,614.2	−28,583.4
Gaussian HMM, K=3	14,323.8	14	−28,619.7	−28,533.4
Gaussian HMM, K=2	14,207.0	7	−28,400.0	−28,356.8
GARCH(1,1)-Normal	13,867.3	4	−27,726.7	−27,702.0
Gaussian i.i.d.	13,563.8	2	−27,123.5	−27,111.2

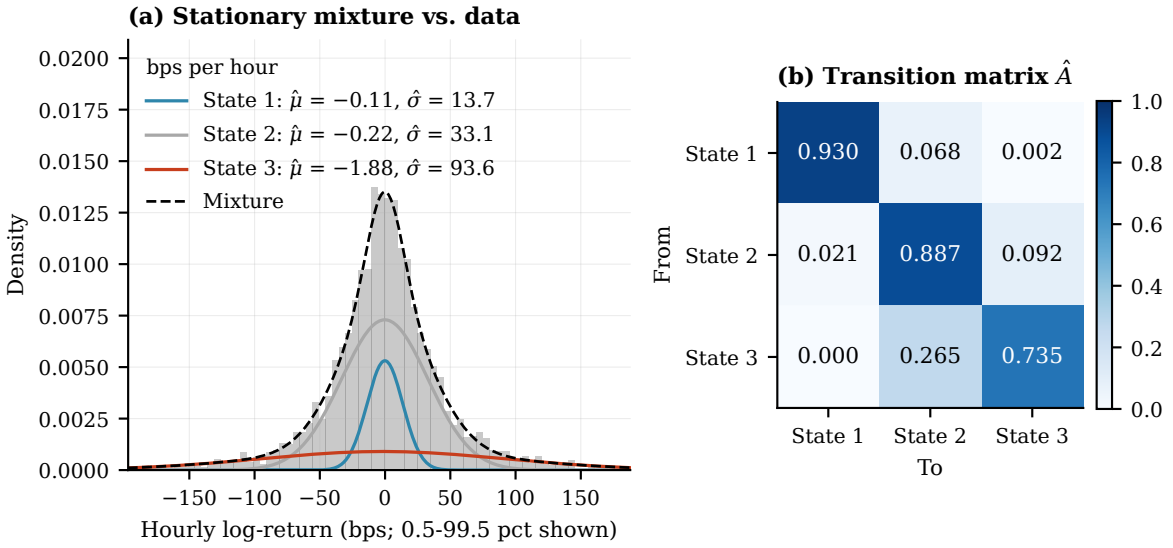


Figure C1. The three-state Gaussian HMM on the hourly sub-period ($T = 3{,}507$). **(a)** Empirical density of hourly log-returns (grey) against the stationary mixture, cut at the 0.5th and 99.5th percentiles. **(b)** The estimated kernel $\hat{A}$, with the high-to-low zero of Section 6.2 at $(3, 1)$.

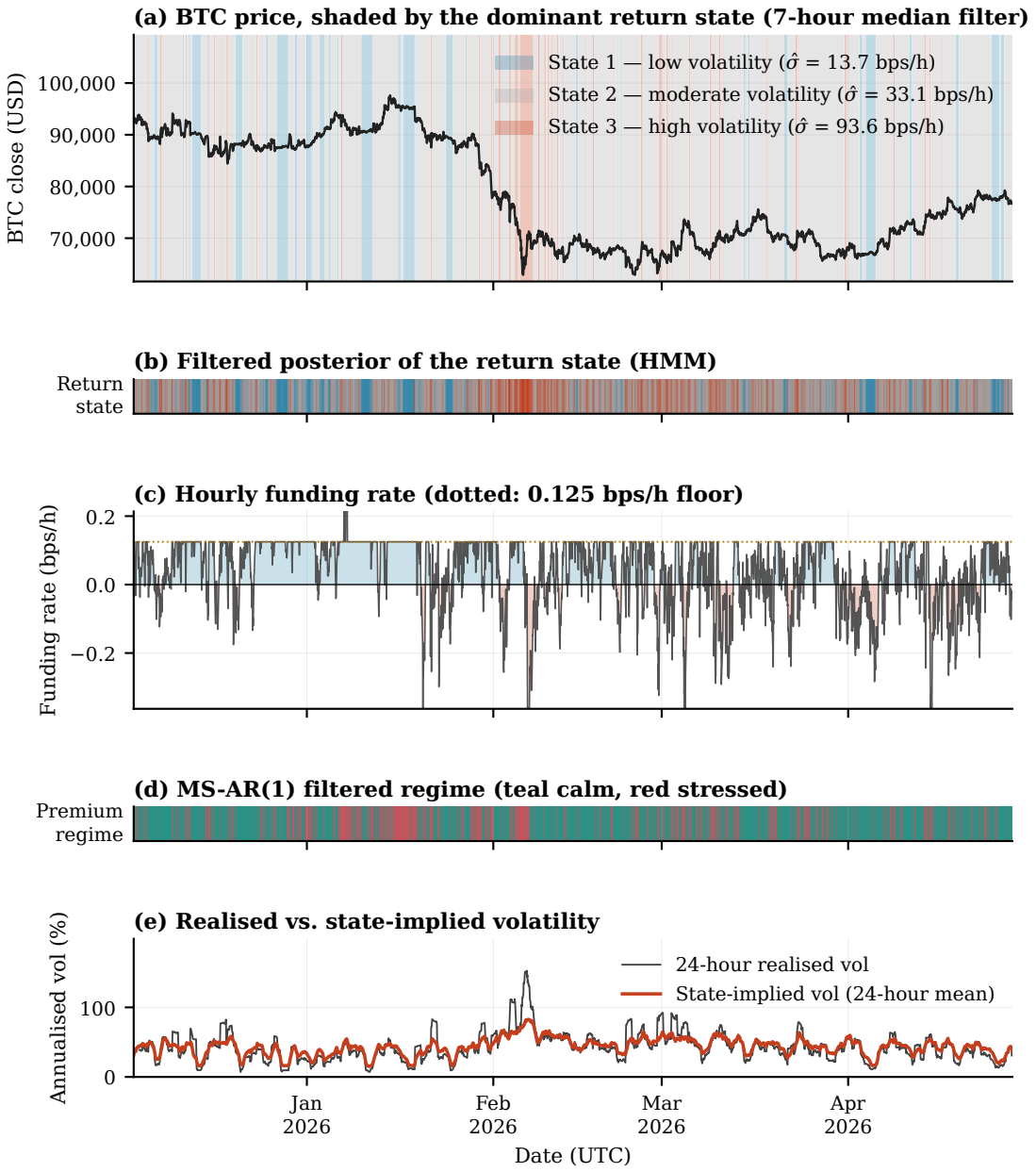


Figure C2. Filtered posteriors over the hourly sub-period, 3 December 2025 to 28 April 2026. **(a)** Closing price, shaded by the dominant return state after a centred seven-hour median filter, which is non-causal and presentational only. **(b)** Filtered posterior of the three-state HMM as mixed colour, blue, grey and red for the low-, moderate- and high-volatility states. **(c)** Hourly funding rate; blue and red fills mark rates above and below zero, the dotted line is the floor, and the range is cut at the 0.5th and 99.5th percentiles. **(d)** Filtered posterior of the MS-AR on the premium re-estimated on this window, where the half-life ordering is reversed; it is not the fit of Table 6. **(e)** 24-hour realised volatility against the 24-hour trailing mean of the state-implied conditional volatility, both annualised.